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Evaluation of millimeter wave radar-based
smart home monitoring systems in health
care applications for elderly people.

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ABSTRACT

Smart home solutions to support the healthcare of older adults and individuals with disabilities demand reliable and non-invasive sensors. Millimeter Wave ([mmWave](#)) radar offers a promising alternative to traditional sensors due to its privacy-preserving nature and resilience to environmental conditions. This thesis evaluates the feasibility of mmWave radar-based systems for healthcare-oriented smart home monitoring. This includes evaluating mmWave radar technology and the systems implemented for distinct applications for smart home-based healthcare. The study was conducted in the Halmstad Intelligent Home ([HINT](#)) environment using the Texas Instruments ([TI](#)) IWR6843ISK radar evaluation board. The developed systems targeted functional, safety, security, social, and physiological health monitoring categories, implementing applications for localization, human activity recognition, fall detection, intruder detection, visitor detection and identification, and heart and breath rate measurement.

An updated benchmark model was adopted to comprehensively evaluate the implemented radar-based systems, focusing on sensor features and system-level performance. While demonstrating the potential of the selected mmWave radar, the results highlighted certain limitations. For instance, accurate heart and breath rate monitoring requires specific positioning and deeper chest movements. Applications like human activity recognition and visitor identification, which use deep learning models, require standardization in data collection and have generalization issues. Additionally, the limited field of view of the radar system hinders coverage, requiring multiple units for larger environments. The accuracy of safety systems like fall detection can be improved by implementing better algorithms or using generalized machine-learning techniques. The study also highlights difficulties distinguishing between multiple individuals, mainly when they are close to each other. Additionally, the accuracy of all the systems was less near the boundaries of the radar's field of view.

Hence, this study highlights that mmWave radar technology is feasible for smart home healthcare monitoring while identifying areas requiring further research and development. Future advancements in radar technology and data analysis techniques could address these limitations, paving the way for more robust and comprehensive solutions.

Keywords: Radar, mmWave, smart home, health monitoring, functional, security, social, safety, physiological, point cloud, track, benchmark score, Convolutional Neural Network ([CNN](#)), Long Short Term Memory Architecture ([LSTM](#))

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ACRONYMS

PIR	Passive Infrared
EM	Electromagnetic
RADAR	Radio Detection and Ranging
ADC	Analog to Digital Converter
CW	Continuous Wave
FMCW	Frequency Modulated Continuous Wave
IF	Intermediate Frequency
AoA	Angle of Arrival
mmWave	Millimeter Wave
HAR	Human Activity Recognition
CNN	Convolutional Neural Network
LSTM	Long Short Term Memory Architecture
MD	Micro Doppler
FFT	Fast Fourier Transform
RSS	Receiver Signal Strength
HINT	Halmstad Intelligent Home
TI	Texas Instruments
FoV	Field of View
CSV	Comma-Separated Values
MIMO	Multiple Input Multiple Output
SNR	Signal to Noise Ratio
EvAAL	Evaluation of Ambient Assisted Living
TLV	Type Length Values
RCS	Radar Cross Section

INTRODUCTION

Smart home systems integrate various network-enabled sensor technologies to monitor events taking place within the home. These systems can learn the routines and preferences of the residents and take actions autonomously or with minimal intervention from the residents [4], [5]. The level of autonomy these systems exhibit defines their degree of intelligence or "smarts" [6].

Over the past decade, there has been increased interest in using smart home technologies for healthcare purposes, particularly to monitor and assist older and disabled people living alone [7]. Smart homes have the potential to recognize people's activities and locations and detect problems, such as older people falling. In the context of a smart home environment, monitoring and assistance play a pivotal role in ensuring the well-being and safety of residents while promoting independent living [8].

Significant application areas of smart homes in the healthcare domain include rehabilitation, Ambient Assisted Living (AAL) [9], continuous health and activity monitoring, and response to abnormal situations [7]. Health-related smart home technologies can be classified into six categories[7]:

- Physiological monitoring focuses on tracking vital signs and health indicators, providing insights into resident's overall well-being.
- Functional monitoring detects activities of daily living and enables timely interventions and assistance when needed.
- Safety monitoring and assistance identifies environmental hazards like fire and gas leaks within the home and assists in preventing accidents or injuries.
- Security monitoring includes the detection of human intruders, ensuring timely responses to the threats.
- Social interaction and assistance monitoring tracks communication and activity participation, fostering social engagement and connection.
- Cognitive and sensory assistance includes reminding intake of medicines, locating lost keys, and technologies that aid sight, hearing, and touch.

Smart homes integrate various ambient and wearable sensing technologies to monitor occupants and the environment to measure physical activity, motion, contact, and presence [10]. Traditional sensor technologies used in smart homes include cameras, smartwatches, microphones, ultrasonic motion sensors, and passive infrared sensors. The challenges associated with wearable sensors, such as smartwatches, include the potential for users to forget to wear them or be reluctant to do so consistently [11], as well as the need for regular removal for charging. Concerning ambient sensors, cameras may prove inefficient in low-light conditions and raise privacy concerns due to their intrusive nature. Passive Infrared (PIR) sensors, for instance, may struggle to detect minor human movements or when the target remains stationary for extended periods [12]. Ultrasonic and acoustic audio sensors are prone to interference from background noise, which can lead to inaccuracies in sensing. Recently, radar technology has emerged as a promising approach for environmental and occupant monitoring.

1.1 PROBLEM DEFINITION

Compared to traditional smart home sensors, radar sensors offer several advantages. Radar sensors can unobtrusively capture the context of daily living activities while being contactless [13]. Visually identifiable data are not provided by radars, making it suitable for placement in sensitive areas like bedrooms or bathrooms [14].

Ensuring reliable and continuous monitoring within a smart home environment while respecting residents' privacy and autonomy presents significant challenges [8]. Furthermore, rigorous testing within a real-world environment is essential to evaluate the feasibility of radar technology in transforming smart home healthcare monitoring. To understand the implications and feasibility of radar-based smart home monitoring systems, the implemented radar-based systems need to be evaluated. Consequently, appropriate evaluation methodologies are crucial for assessing radar systems' capabilities in this domain.

Braun et al. [3] introduce a benchmark model for evaluating sensors within smart environments. This model considers sensor features like resolution, update rate, and detection range and assigns a weight based on the application requirements. By comparing sensor scores across different applications, this model aids in sensor selection.

J. Antonio et al. [15] propose an evaluation method for Ambient Assisted Living systems based on performance metrics, including accuracy, response time, availability, and installation complexity. The model offers practical implications for assessing smart home systems.

Hence, both evaluation methods can be combined to comprehensively assess the sensor and sensor-based system for enhanced feasibility studies. However, for the benchmark model proposed by Braun,

the weights assigned for each application are subjective to the authors, and the same is followed in the combined model. To back the ratings given by the authors, an online survey is conducted to get better insights into finalizing the weights for each application.

This thesis investigates the potential of single-radar systems for diverse health-related monitoring applications within a smart home environment. By evaluating the capabilities of radar technology and conducting real-world experiments, this research aims to assess the feasibility, effectiveness, and implications of integrating radar systems into healthcare monitoring solutions.

1.2 NOVELTY

This thesis assesses the feasibility of employing radar systems for different categories of smart home health-related monitoring applications.

Unlike previous research focusing on specific monitoring categories, this study investigates the comprehensive application of radar technology across various health-related domains within a real-world environment. This thesis implements radar-based systems to monitor functional, safety, security, social, and physiological aspects of an elderly individual's life within a smart home environment. Functional monitoring encompasses resident tracking and activity recognition. Safety monitoring prioritizes fall detection. Security surveillance focuses on intruder detection during absence. Social monitoring involves visitor detection and recognition. Lastly, physiological monitoring tracks heart and breath rates.

By combining elements from the sensor evaluation benchmark model proposed by Braun et al. [3] with additional metrics inspired by J. Antonio et al.'s [15] evaluation method for AAL systems, this thesis offers a novel approach to assess radar systems for smart home healthcare applications.

Based on the evaluation, the thesis also discusses the implications of using radar-based smart home monitoring systems for healthcare applications.

1.3 OUTLINE

The remainder of this thesis is structured as follows: Chapter 2 establishes the context underlying radar systems, covering their basic principles, types, and components. This overview will set the stage for understanding radar's operation and significance in various contexts. Chapter 3 reviews existing research and literature concerning radar-based systems in functional, safety, security, and social monitoring. Chapter 4 details the methodology employed. Achieved results

are presented in Chapter 5 and discussed in Chapter 6. Chapter 7 presents the main conclusions.

BACKGROUND

This chapter covers the fundamentals of radar systems, encompassing various principles and types of radar. This chapter explores the applications, advantages, and disadvantages of using radar systems in smart home environments.

2.1 FUNDAMENTALS OF RADAR SYSTEMS

Radio Detection and Ranging (**RADAR**), utilizes Electromagnetic (**EM**) waves transmitted towards an area of interest to detect objects. As shown in Figure 1, the transmitter antenna emits EM waves, which are then reflected by targets in the region of interest and received by the radar system's receiver antennas. After reception, the Analog to Digital Converter (**ADC**) converts the received signals into digital format for processing. Finally, the signal processing unit extracts information from the digitized signals [16].

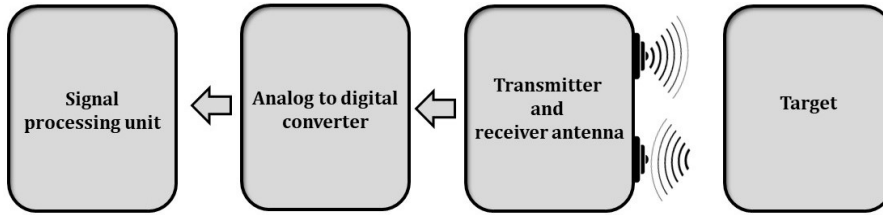


Figure 1: Radar working principle.

The total distance travelled by the EM wave is $2R$ (R is the range to the detected target) as it travels to the target and back. The speed of an EM wave is the speed of light ($c = 3 \times 10^8$), and the distance is speed multiplied by the time. The equation for computing the range can be seen in Equation 1 [17].

$$R = \frac{c\Delta T}{2} \quad (1)$$

2.1.1 Types of radar

Radars can be classified into pulsed, continuous wave, and Frequency Modulated Continuous Wave based on the fundamental principles or techniques used in radar systems [16].

2.1.1.1 Pulsed radar

The pulsed radar emits short bursts of electromagnetic energy followed by periods of inactivity, during which the radar receiver listens for echoes from objects in its field [16]. The time it takes for these echoes to return provides information on the distance to the target. However, the pulsed radar faces difficulty distinguishing desired echo signals from unwanted leakage signals. This difficulty arises due to the lower power of received echo signals than transmitted power. Although separate antennas for transmission and reception can aid in segregating signals, the achieved isolation is often insufficient.

2.1.1.2 Continuous wave radars

In contrast, Continuous Wave (CW) radars emit a continuous stream of electromagnetic waves [16]. A CW radar can detect the speed and direction of moving objects by analyzing frequency shifts in the reflected waves due to the Doppler effect. This effect is caused by relative motion between the radar and the target. The challenge with CW radar lies in its narrow spectrum; a timing mark is necessary during transmission to determine the range accurately. However, a sharper timing mark leads to a broader transmitted spectrum. Amplitude, frequency, or phase modulation techniques are applied to broaden the transmission spectrum and facilitate range measurements.

2.1.1.3 Frequency modulated continuous wave radar

Frequency modulation is a commonly used method for spectrum broadening, where the changing frequency acts as the timing mark [16]. The transit time is proportional to the difference between the frequency of the echo and transmitter signals. Increasing the frequency deviation over time enhances the accuracy of transit time measurement and widens the transmitted spectrum. This method, known as Frequency Modulated Continuous Wave (FMCW), addresses the limitations of CW radar by utilizing frequency modulation to broaden the spectrum and enable accurate range measurement. This signal, in which the frequency increases or decreases linearly over time, is called a chirp signal. Figure 2 illustrates a chirp signal in the time domain with linearly increasing frequency.

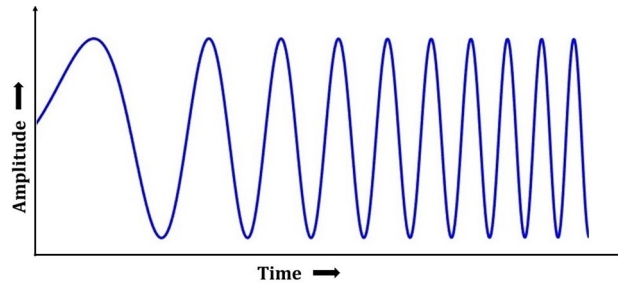


Figure 2: Chirp signal with amplitude as a function of time.

Figure 3 illustrates the chirp signal in the frequency domain. A chirp is characterized by its start frequency (f_c), bandwidth (B), and duration (T_c). The chirp's slope (S) represents the frequency change rate. In the example shown in Figure 3, the parameters are $f_c = 60\text{GHz}$, $B = 4\text{GHz}$, $T_c = 40\mu\text{s}$, and $S = 100\text{MHz}/\mu\text{s}$.

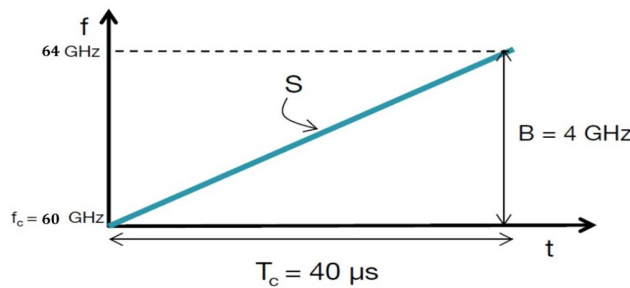


Figure 3: Chirp signal, with frequency as a function of time (adapted from [1]).

Figure 4 represents a simplified block diagram of the main RF components of an FMCW radar. It includes a synthesizer, a mixer, and the transmitter and receiver antennas.

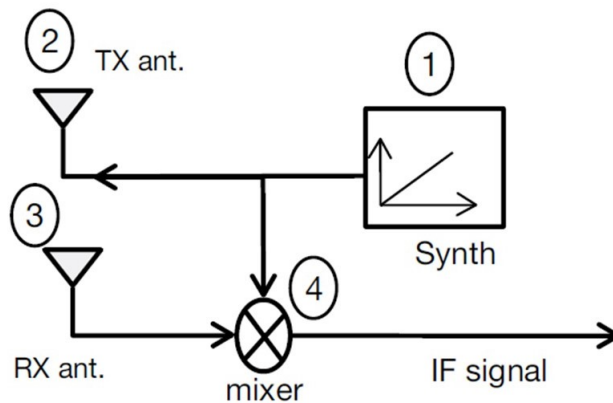


Figure 4: FMCW block diagram [1].

The following steps explain the working of the block diagram [1] illustrated in Figure 4.

- The synthesizer generates a chirp.
- The chirp is transmitted by the transmit antenna (TX ant).
- The reflection of the chirp by an object is captured by the receive antenna (RX ant).
- A mixer combines the RX and TX signals to produce an intermediate frequency (IF) signal.

In FMCW radars, a synthesizer generates chirp signals, characterized by a sinusoidal waveform with a frequency that increases linearly over time. The transmitting antenna transmits the chirp, and the receiving antenna receives a time-delayed version after hitting an obstacle. The transmitted and the received signals are mixed to form the Intermediate Frequency (IF) signal. The frequency of the IF signal is directly proportional to the target's distance [18].

The upper part in Figure 5 illustrates TX and RX chirps over time for a single detected target. Notably, the RX chirp is a time-delayed version of the TX chirp. This time delay (t) can be mathematically derived as shown in Equation 2 where d represents the distance to the detected object and c stands for the speed of light[1]:

$$t = \frac{2d}{c} \quad (2)$$

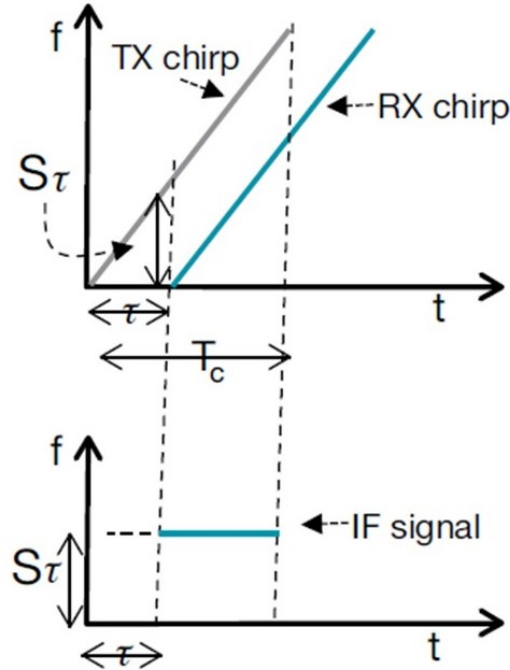


Figure 5: IF signal [1].

To acquire the frequency representation over time of the IF signal at the output of the frequency mixer, the two curves depicted in the upper section of Figure 5 are subtracted. The distance between these two curves remains fixed, indicating that the IF signal consists of a tone with a constant frequency. Figure 5 also illustrates that this frequency is denoted as S_τ . Notably, the IF signal remains valid within the time interval where both the TX and RX chirp overlap, i.e., the interval delimited by the vertical dotted lines in Figure 5 [1].

As a function of time, the mixer output signal manifests as a sine wave owing to its constant frequency. The initial phase of the IF signal (Φ_0) is determined by the disparity between the phase of the TX chirp and the phase of the RX chirp at the precise moment corresponding to the initiation of the IF signal. This relationship can be mathematically expressed as Equation 3 [1]:

$$\Phi_0 = 2\pi f_c \tau \quad (3)$$

where τ represents the time delay.

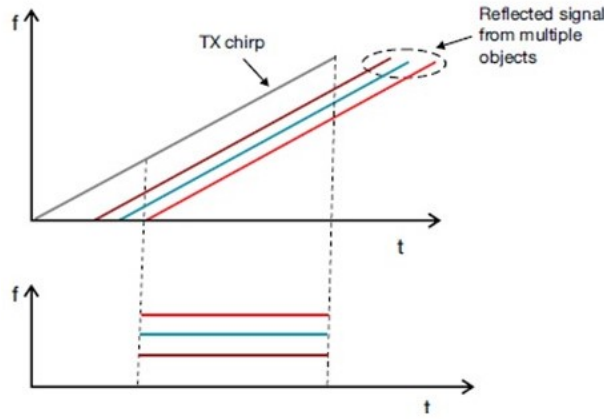


Figure 6: Multiple IF signals [1].

A similar observation holds true if multiple moving objects are in the vicinity, as illustrated in Figure 6. This IF signal, which consists of multiple tones, must be processed using a Fourier transform to separate the different tones. Fourier transform processing will result in a frequency spectrum and the spectrum's peak denoting an object's presence at a specific distance. This spectrum of frequencies is known as beat frequencies [1]. The beat frequency (f_{beat}) is directly proportional to the difference in frequencies between the transmitted and received signals, which in turn is proportional to the time delay (τ) between the transmission and reception of radar pulses. Mathematically, this relationship is expressed as in Equation 4

$$f_{\text{beat}} = S \cdot \tau \quad (4)$$

Where S is the frequency modulation rate, the beat frequencies represent the various time delays between the radar pulses and are utilized to accurately determine the range or distance to the detected targets. The IF signal contains multiple sinusoidal components in scenarios with multiple targets, each corresponding to a specific target. Analyzing the spectrum of beat frequencies enables the radar system to distinguish between multiple targets based on their respective distances from the radar.

The ability to distinguish different targets that are closely spaced is referred to as range resolution [18]. The bandwidth of the chirp signal and the speed of light determines the range resolution. Equation 5 shows this relationship, where ΔR is the range resolution, c_0 is the speed of light, and B is the bandwidth.

$$\Delta R = \frac{c_0}{2 \cdot B} \quad (5)$$

Hence, a large bandwidth gives less ΔR that enables the measurement of small distances in the proximity of the sensor.

The speed or velocity of a target is determined by analyzing the target's radial velocity and by examining the change in phase of the IF signal from one chirp to another [1]. The phase difference between consecutive chirps is proportional to the target's radial velocity, as shown in Figure 7. However, a constraint exists on the phase difference due to its periodic nature, which sets an upper limit on the detectable velocity. This constraint necessitates adjustments in parameters such as chirp duration or frequency to improve the radar's ability to measure higher velocities.

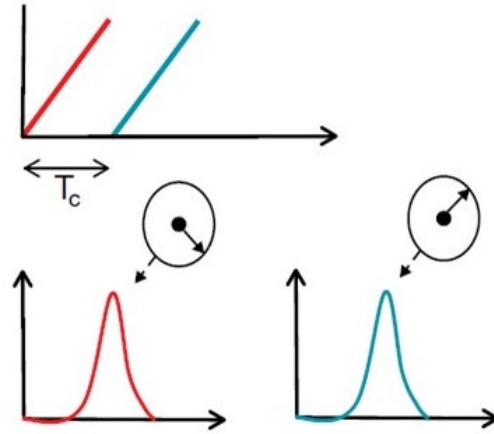


Figure 7: Velocity measurement [1].

An FMCW radar system can estimate the angle of a reflected signal from a target, as shown in Figure 8.

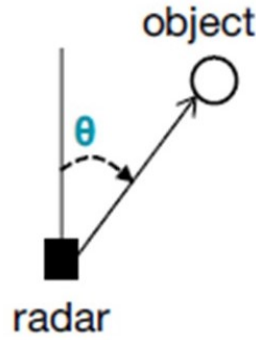


Figure 8: Angle of arrival [1].

This angle is commonly referred to as the Angle of Arrival (AoA) [1]. Angular estimation relies on the observation that a slight alteration in the distance of an object induces a phase shift in the peak of the range-FFT. This phenomenon is utilized for angular estimation, employing at least two RX antennas as illustrated in Figure 9.

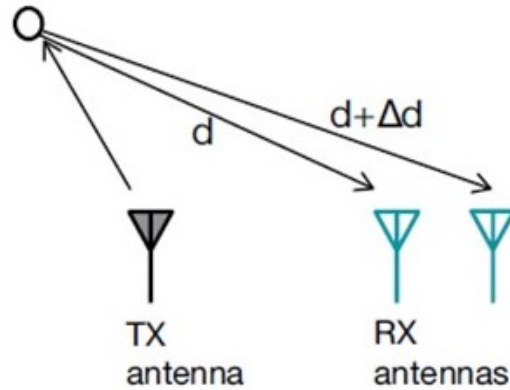


Figure 9: Multiple antenna to estimate AoA [1].

The variation in distance from the object to each antenna leads to a phase shift in the Fast Fourier Transform (FFT) peak; this phase differences between signals received by different antennas are used to compute the angle of arrival of the detected object relative to the radar, providing valuable spatial information about the targets.

mmWave radars leverage the foundational concepts of FMCW and have gained popularity in a wide range of applications. The following section discusses mmWave radars in more detail.

2.2 MMWAVE RADAR

Traditional radars generally operate in the range of 220 MHz to 35 GHz. Although, these are not a fixed range as radars can operate in ranges above and below these frequencies [16]. mmWave radars oper-

ate at higher frequencies, and the most common industrial mmWave radars are in the 24–80 GHz frequency range. Due to high operating frequency and large bandwidth, mmWave radars can provide high precision, as explained by the equation 5. The size of the antennas in the radars is proportional to the wavelength. The millimetre wavelength in mmWave radars enables the use of small antennas, which led to the production of miniature-sized radar systems [19]. The characteristics of the high-frequency electromagnetic spectrum of mmWave make it resistant to weather and illumination. It has a wide sensing range and non-invasive sensing capabilities [20].

Increasing the number of antennas in mmWave radar systems enhances spatial resolution, allowing for the precise detection of targets and the differentiation of closely spaced objects. This improvement is vital for applications requiring accurate detection and tracking [21]. Although increasing the number of elements in the antenna array improves the sensor's resolution, it raises the overall sensor dimension and complexity of the system. Multiple Input Multiple Output (MIMO) systems are an alternative solution for replacing large phased arrays[22]. MIMO radar transmits multiple probing signals, which creates a virtual aperture wider than the actual physical aperture, resulting in improved angular resolution. The effectiveness of this virtual array depends not only on the transmitted waveforms but also on the physical arrangement of the transmitting and receiving elements. MIMO antenna systems are used in mmWave radars, which increases the detection resolution. These characteristics of mmWave radars make them well-suited for various indoor applications.

2.3 APPLICATIONS OF RADAR-BASED SYSTEMS

Radar technology was primarily applied in the military and aviation sectors, serving essential roles in surveillance, navigation, and defense [16]. The mmWave technology allows for tiny antennas, creating compact sensors. These sensors can be conveniently installed in various devices and locations, including behind car bumpers, security cameras, or inside drones and robots. This flexibility makes it possible to deploy these sensors effectively for tasks like environmental monitoring and traffic control [19]. Hence, in recent years, radar systems have been used significantly in the automotive industry due to the sensors being compact and affordable [23]. The compactness and the reduction in the cost of radar sensors have led to their usage in indoor applications and adoption into healthcare systems and an increase in research in smart home monitoring systems [24], [25].

LITERATURE REVIEW

The combination of miniaturized mmWave radars and the FMCW technique, which offers advantages like continuous monitoring and high sensitivity, enables their use in smart home healthcare applications, allowing for accurate, non-invasive, and non-obtrusive detection and monitoring of people. This chapter discusses the related works of radar systems focusing on functional, security, safety, social activity, and physiological monitoring (Figure 10). Possible evaluation methods for using sensors for different applications in a smart environment are also discussed in this chapter.

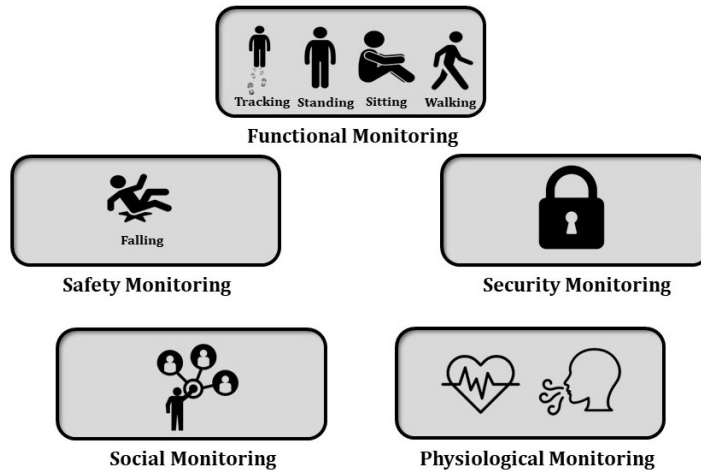


Figure 10: Functional monitoring includes detecting and tracking daily routines. Safety monitoring prioritizes the resident's well-being, such as detecting falls. Social monitoring assesses the frequency of a resident's social interactions. Security monitoring caters to potential security threats by detecting intruders. Physiological monitoring includes heart and breath rate detection.

3.1 RADAR-BASED FUNCTIONAL MONITORING

In the context of smart home healthcare applications, functional monitoring is the collection and analysis of the daily activities of the residents [7]. This monitoring involves the detection and tracking of movements and activity recognition of the smart home resident. This section reviews existing literature on the use of radars, focusing on their application in functional monitoring. It explores the research on the impact of sensor positioning, signal processing algorithms, and radar sensor platforms for detecting human presence and tracking

movement within an indoor environment. It also discusses the related works of using radar for Human Activity Recognition (HAR).

Radar placement varies across studies, with some opting for ceiling-mounted sensors [26], [27] while others utilize multiple wall-mounted units [28], [29]. While ceiling placement often enhances detection accuracy, side wall placement can broaden coverage.

Pirosa [30] has done a detailed study and comparison of radar sensors for indoor tracking. The two sensor platforms used for comparison are Infineon's Position2Go and TI IWR6843ISK. Position2Go is a 24 GHz radar board, whereas IWR6843ISK is a 60 GHz radar board, and both are interfaced with an ARM-based microcontroller. The experiments use the inbuilt firmware provided by the suppliers, which supports tracking multiple targets. The experiments took place in laboratory, corridor, and terrace environments. The laboratory environment is surrounded by tables and chairs, imitating an actual room in a smart home. The corridor is an empty large area surrounded by walls, and the terrace only has railings. The two sensors were kept at a fixed location in each environment, and a fixed path was taken to track human movements. A fixed path is decided for a person to walk on in the three environments, and the readings from both sensors are taken while the person repeatedly walks on the fixed path in all three environments. The evaluation methods used by the author to compare the sensors are area error, regression error, acceleration error, and relative location error. The evaluation results showed that the original movement's relative location error of IWR6843ISK is much lower than Position2GO. The other evaluation methods were similar for both sensors. Position2Go radar often loses track of the detected target compared to the IWR6843ISK sensor. In overview, the IWR6843ISK sensor performed better in tracking than Position2GO.

The distance between the reference and measured points is considered a standard metric discussed across various studies and is used to assess the feasibility of radar-based localization systems [31, 32, 30]. The first approach shows that the mmWave radar-based localization systems have achieved a minimum mean error of 0.6974 m [31]. In the second method, the authors claim that their indoor localization system based on mmWave radars is feasible with an estimated position error of 0.42 m [32]. In the third study, the range of position errors for the TI IWR6843ISK in a lab environment across different configurations spans from 1.048 m to 0.134 m [30].

HAR uses radar to analyze radar signals to detect and classify human activities such as walking, running, and sitting. By analyzing Doppler shifts, signal strength, and time-frequency characteristics, radar systems can distinguish between human activities.

The primary challenge in the HAR system is the differences in the users, leading to differences in their actions based on individual habits and physical characteristics [13]. Additionally, these habits can

change over time, which may also affect the system. Moreover, the sensor placement can also affect the echo signals, impacting feature extraction. Furthermore, differences in environmental layout from room to room can yield varying results. A robust radar-based HAR system can overcome these external factors and maintain reliability across different conditions.

Leo and Tim conducted a study on using an algorithm to recognize basic human activities with mmWave radars [33]. The sensor used for the experiment is AWR6843AOP by TI, which operates at 60 GHz frequency. The pre-existing software tool provided by TI called the area scanner, was used in the experiment. This tool helps provide information about human detection with the data coming from the sensor. This data is used to classify simple human activities like standing, sitting, lying down, and falling. The sensor is kept on a tripod at a height of 2.36 meters and tilted to an angle of 30 degrees downwards. The authors have considered an average of the first 100 frames as the reference height. Then, the average of the target's height for ten frames is taken and compared with the reference height. If the average height is more than 75% of the standard height, it is considered that the human is standing; if the average height is between 50 – 75 %, it is considered that the human is sitting; and if it is less than 50%, then it is considered that the human is lying down. If the activity state changes from standing to lying down quickly, it is assumed to be a fall. The results of the experiments showed that the accuracy for standing, sitting, lying, and falling are 62.2%, 52.7%, 23.6%, and 73.5%, respectively.

Introducing Deep learning models has improved the ways to extract complex human activities, allowing for more accurate and robust recognition algorithms [34].

A CNN, along with a LSTM, is proposed for HAR [35]. A single FMCW radar operating on 60GHz is used for collecting data. Data were collected in three rooms with varying clutters; two rooms with the least clutter were used for training, and the other was used for validation. The time-range, time-doppler, and range-doppler projections of the received radar data are given to the proposed CNN-LSTM architecture for feature extraction and then the classification of a few activities like walking towards and walking away from the sensor, sitting down, and standing up. The highest accuracy received for the classes walking away, walking towards, sitting down, and standing up are 94%, 87%, 63%, and 58%, respectively. Thus, the total average accuracy of the model in an unseen environment is 75.25%. This might be due to the clutter in the different rooms and the different viewing angles of the radar.

An end-to-end CNN architecture that learns signal representation, feature extraction, and classification of HAR from raw radar data is proposed [13]. Two 1-D convolution layers are added to learn the

radar signal representation. These layers replace the short-term Fourier transform layers used to represent raw radar data in the time-frequency domain [36]. After that, a traditional Deep CNN is used to classify seven activities: walking, walking while holding a stick, running, crawling, sitting still, boxing at a place, and boxing while moving forward. For running, 3 seconds of data were recorded, and for the rest of the activities, 6 seconds were recorded. Boxing while standing and sitting still activities were collected at a distance of 0.5 meters from the radar, and the remaining activities were recorded at 5 meters from the radar. The overall average accuracy on the test data set is 96.3%, in which the accuracy of the activities sitting still and walking are 100% and 94%. Although the results are better compared to other models, the computation is higher for this model due to the increase in hyperparameters, and the other limitation is that the model may overfit.

In the case of older adults living alone, the localization details coupled with essential human activity recognition like walking, sitting, and standing give a general pattern of the daily living activities within a smart home. This can be used to monitor the functional parameters of the resident.

3.2 RADAR-BASED SAFETY MONITORING

In the context of smart home healthcare applications, safety monitoring detects and identifies environmental hazards and accidents [7]. According to L.Z Rubenstein [37], 40% of people older than 65 fall at least once a year in their homes. The older adults who fall take time to get up [38], and a corresponding study shows that the risk of death almost doubles for older adults who are lying on the floor for an extended period of time after a fall [39]. Given the high incidence of falls among elderly individuals and the significant increase in mortality risk associated with prolonged time spent on the floor after a fall, safety monitoring systems must detect falls in a timely manner to ensure timely assistance. Radar sensors can detect accidental falls in a home and ensure the resident's safety. Recent work that used radar for fall detection is discussed below.

An approach for fall detection using mmWave radars was presented by Wang et al. [40]. The authors used four different methods to estimate the person's body height. A fall is detected if the person's height drops beyond a certain threshold. The experiments were conducted in three different environments to test the algorithm's feasibility. The overall accuracy achieved for the system was 82.82 %.

The HAR method discussed earlier also detected falls [33]. If the activity state changes from standing to lying down quickly, it is assumed to be a fall. The accuracy for fall was 73.5%.

Another approach used a combination of two mmWave radars with a bidirectional LSTM model to detect falls [41]. The authors tried

different combinations of radar placements and achieved an accuracy ranging from 90% - 98%.

Since radars cannot be used to detect hazards like fire or gas leaks, this thesis limits the safety monitoring to detecting accidental falls in the smart home.

3.3 RADAR-BASED SECURITY MONITORING

In the context of smart home healthcare applications, security monitoring tries to detect intruders [7]. Radar sensors can be used to detect intruders and provide security to the residents.

Okar and Wu [42] propose mmWave radars as a viable alternative to traditional video surveillance for the International Atomic Energy Agency (IAEA) in nuclear material storage areas, particularly in scenarios where video systems might struggle, such as in low visibility or during power outages. They utilized Texas Instruments' IWR6843ISK mmWave sensors, detecting multiple intrusions even under these challenging conditions. The study also underscores the need for optimal sensor placement to minimize blind spots and suggests further exploring additional mmWave sensors to enhance monitoring capabilities.

In another study, Vandana G. S. et al. [43] explore the application of mmWave radars, specifically the IWR1642 FMCW radar sensor, for intruder detection and tracking. They intended to evaluate the effectiveness of radar in tracking moving targets of various Radar Cross Section (RCS) values, such as runners, cyclists, and motorcyclists. The study concludes that radar-based tracking was most reliable for targets with larger RCS values, but camera-based tracking was more effective for mid-range RCS targets like cyclists. The results demonstrate that mmWave radars are a viable tool for intruder detection, with future work suggested to improve accuracy over longer ranges and in more complex scenarios.

3.4 RADAR-BASED SOCIAL MONITORING

In the context of smart home healthcare applications, social monitoring detects the residents' social activities, including phone calls, visitors, and other virtual activities [7]. This section discusses the related work that monitors the social life of residents in a home environment, the limitations of existing technology, and how radars can be used to monitor social activities.

The detailed study reveals a positive association between social connectivity and the health status of older adults [44]. This finding shows the importance of fostering social connections for the well-being of elderly individuals. Recent advancements in smart home technologies offer promising avenues to enhance or sustain the social connectivity

of older people residing alone in their homes [45]. A method called Visitrack, proposed by Gaete et al. [46], utilizes a tablet's frontal camera and microphone to monitor the social activity of older adults living alone. If the sound captured by the tablet's microphone is above a threshold value, the system switches on the camera and takes a picture. The system detects faces, crops them, and sends them to a server for identification. The server creates and maintains records of the social visits based on the database in the server. This method showed promising results in detecting the time spent with the visitors when they are in the line of sight of the camera. The limitation of this method is that it uses a camera from a tablet, and if the camera is not in the line of sight of the resident or the visitor, it will fail to detect. Using cameras for monitoring is also considered obtrusive. Microphones can pick up environmental noises, such as when a television or radio is playing. Radars offer a non-obtrusive way of monitoring and are also resistant to environmental noises such as sound and light.

Multiple human detection is possible with the advancements in radar technology as discussed in section 3.1. This can be utilized to monitor the social life of older adults living alone. Gait recognition of radar data can be used to identify frequent visitors and keep track of them instead of using cameras.

In one approach, Micro Doppler (MD) signature is processed from the raw data from the radar and used to identify the gait of different individuals [47]. The MD signature is fed to a deep CNN identification network to identify known users and intruders. The radar used to collect data is TI's IWR1443 EVM, which operates at 77 GHz. The setup is done in an indoor corridor, and the system has an accuracy of 88.59% in identifying five subjects.

A gait recognition system named MTPGait is used to extract spatiotemporal features from radar point cloud data [48]. The proposed method uses a neural network with CNN combined with LSTM, followed by a fully connected layer. The experiments were done in three rooms: a laboratory, a conference room, and a corridor. These have varying sizes and sensing areas. Nine hundred sixty minutes of data were collected while the radar was one meter above the ground. The processed radar data has X, Y, Z, V, and N where X, Y, and Z are the point coordinates, V is the Doppler velocity of the point, and N is the Signal to Noise Ratio (SNR). The number of points in a frame keeps varying, so 128 points were fixed for a frame to make it uniform. If there were fewer points than the fixed threshold, additional copies were made to reach 128. For model training, 30 frames were used as input. Forty volunteers from the age group of 19 and 29 were used to collect data. The system was able to achieve 96.7% recognition accuracy when a single person was there in the sensing area and 90.2% in a multiple-person scenario. Using radar point cloud data shows

better results in classifying individuals with gait patterns. A deep learning model with CNN combined with LSTM can be utilized.

This thesis concentrates on the in-person social activities of the residents and identifying the frequent visitors using gait recognition models for social monitoring.

3.5 RADAR-BASED PHYSIOLOGICAL MONITORING

In the context of smart home healthcare applications, physiological monitoring involves the monitoring of heart rate and breath rate. This section discusses the related work that monitors heart and breath rates using radar-based techniques. One of the essential applications of radar-based physiological monitoring is the non-contact and continuous monitoring of vital signs like heart rate and breath rate of humans.

The ability of phase waveform to represent tiny displacements is utilized for retrieving fine-grained vital signs [20]. A phase waveform is formed by extracting the phase values from the Range-Angle spectrum or the Range-FFT of the reflected signals from the mmWave sensor.

A custom-made radar system that uses a 60 GHz mmWave signal continuously monitors human breathing and heart rate [49]. The proposed method transmits the signal to the human body, and the reflected signal's Receiver Signal Strength (RSS) is extracted. The RSS signal's frequency domain gives a peak due to the periodic movement of the human body. These peaks are detected using band-pass filters, resulting in the estimation of breath and heart rates.

An FMCW radar from TI operating in the frequency range of 60-64 GHz is utilized to measure the vital signs of a single target [50]. The signal processing algorithm involves sampling the middle-frequency signal using an ADC, followed by FFT processing in the range dimension to calculate the target distance. Phase information corresponding to the distance unit is then extracted for subsequent algorithm processing. Band-pass filters isolate respiratory and heartbeat bands, and wavelet transform is applied to isolate the heartbeat signal. A similar method is used for vital sign detection of multiple targets utilizing an FMCW mmWave radar operating at 76-81 GHz [51]. Measurements over multiple channels are used to separate different objects by utilizing the range-azimuth plan formed by taking FFT on the range bins. The vital signs of each of the objects are computed by extracting the phase values from the identified range bins.

These are the methods that use frequency domain analysis to extract features. There are also learning-based feature extraction methods [20]. A service robot equipped with commercial mmWave radar is used to detect the heart rate of humans [52]. The robot measures the heart rate by computing the phase variation caused by the micro-

displacement of the user's skin on the leg. This data is mainly filled with noise. The heart rate measurement from this noisy data is formulated as a regression problem and solved using a customized deep neural network. This system is insensitive to the orientation and position of the users.

In recent works mentioned above, the phase information from the reflected radar signal is used to measure the heart and breath rate parameters. These works have utilized customized radar designs to get good results. Designing a new radar system and performing signal processing algorithms to extract phase information is out of the scope of this thesis. The firmware that detects vital signs released by TI for their radar sensors can be utilized to evaluate radar-based heart rate and breath rate monitoring.

3.6 EVALUATION METHODS

This section discusses the related work about using an evaluation method for the sensors in a smart environment.

A benchmark model is proposed to evaluate sensors for different applications in a smart environment [3]. Sensor performance characteristics, pervasive metrics, and environmental characteristics are the different groups considered for the feature matrix of the model.

Once the features and their weights are established, a formal model is applied to calculate a benchmark score for any sensor technology and application.

The author has selected these distinct features such as resolution, update rate, detection range, unobtrusiveness, processing complexity, robustness, disturbance frequency, calibration complexity, and unique limitations for their model. The features are rated in five categories, from least favourable to most favourable, for the sensors to be evaluated. Weights are assigned to each feature based on the importance of that feature to a particular application. The benchmark score is calculated using the sensor feature matrix and the application weight matrix. This benchmark score can be utilized to compare the performance of different sensors for a particular application, or it can be used to understand which application is suitable for a specific sensor. However, the application weights that are assigned for the benchmark scores are subjective to the authors. This is a drawback of this benchmark model.

J. Antonio et al. [15] proposed an evaluation method for localization and activity recognition systems for ambient assisted living. This method uses the following metrics for the evaluation: accuracy, installation complexity, availability or response time, user acceptance, and interoperability. These metrics are normalized to a scale that ranges from 0 to 10, and a weighted average is mixed. The method by which these metrics are computed differs between the two applications. This

method is used as an evaluation method for a competition called Evaluation of Ambient Assisted Living ([EvAAL](#)), which judges the competitors' systems in Ambient Assisted Living. Some of the features of this evaluation method can also be utilized to evaluate smart home monitoring applications.

The combination of the evaluation model for sensors and the method for evaluating the systems can be utilized to study better the feasibility of radar-based systems for smart home monitoring in healthcare applications. Since the application weights are subjective to different authors, a survey can be done to address this limitation. The rating criteria of the new features combined to the existing benchmark model can also be finalized with a survey.

METHODOLOGY

This chapter outlines the methodology for evaluating the feasibility of radar systems in various smart home health monitoring applications. It begins with a detailed description of the smart home environment to establish the context for radar system evaluation. The subsequent sections focus on the selection and operation of the mmWave sensor platform, including firmware details. The evaluation methodology, radar system implementation, and experimental design are then elaborated.

4.1 HALMSTAD INTELLIGENT HOME

Contrasting with previous works, the experiments in this work were conducted in a fully functional apartment named [HINT](#), located on the Halmstad University campus. HINT serves as a laboratory for education and research in smart homes [53]. The apartment comprises six distinct areas: the entrance, living room, bedroom, dining room, bathroom, and kitchen, as depicted in Figure 11.

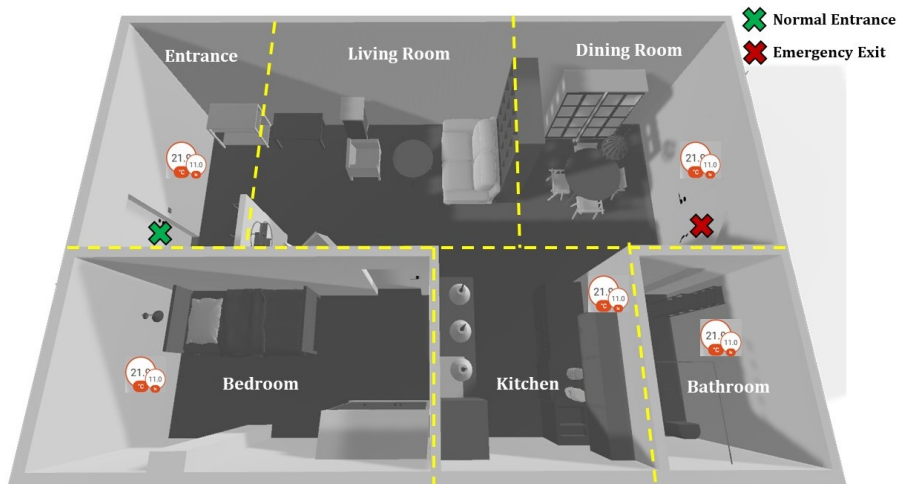


Figure 11: HINT Layout.

The apartment features two entry points: a standard entrance and an emergency exit, as illustrated in Figure 11. It is equipped with modern amenities and includes couches, shelves, a bed, a dining table, chairs, kitchen appliances, and a refrigerator. The bathroom and bedroom are separate from the common living area. HINT incorporates various sensors to measure physical parameters like motion, temperature, light, door/window opening, and occupancy. Au-

tomated lighting control based on sensor data is implemented to enhance functionality.

4.2 MMWAVE RADAR SENSOR PLATFORM SELECTION AND OPERATION

The high detection and angular resolutions of mmWave radars make them suitable for various applications in a smart home. The following discussion outlines the criteria that guided the selection of the specific mmWave radar sensor platform and highlights its key characteristics, providing a foundation for understanding its role in this research.

4.2.1 *Selection criteria*

Informed by the literature review in Chapter 3, the selection of the sensor platform focused on mmWave radars from TI due to their capability to provide range, velocity, and angle information, making them ideal for capturing detailed environmental interactions. TI sensor platforms offer flexibility in operation, allowing users to flash different firmware based on the application, customize existing firmware, or directly access raw ADC data (requiring an additional data capture card) [2].

Building upon Piroso's research [30], which demonstrated the effectiveness of TI's IWR6843ISK in indoor human detection and tracking using the provided firmware, this thesis adopts the IWR6843ISK evaluation board for implementing various smart home health monitoring applications. The board's ability to achieve low relative location error in original track measurements, as detailed in Section 3.1, was a key factor in this decision.

4.2.2 *3D people counting demo firmware*

To demonstrate the TI's mmWave radar capabilities, including the IWR6843, TI provides comprehensive examples with firmware and visualization tools for objects/people detection, gesture recognition, people tracking and counting, and vital signs detection, among others [54]. Among these, the 3D people counting demo is a firmware used to track humans in an indoor environment, and hence, it is used in the experiments conducted for this thesis. This section explains the working and functionalities of the firmware.

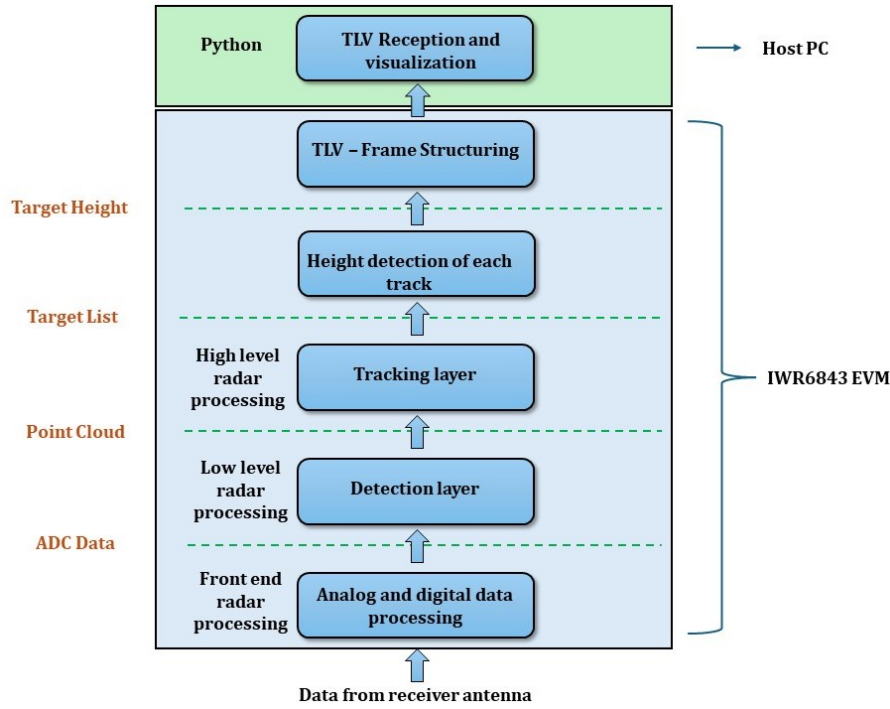


Figure 12: The signal processing layers of the firmware, modified from [2].

As shown in Figure 12, the data from the receiver antenna will go through detection and tracking layers to identify the point cloud and track information. The firmware also provides a height detection algorithm capable of identifying each target's height in its vicinity. The processed point cloud, tracks, and track height are then formatted into specific Type Length Values (TLV) structures before being transmitted serially to the host PC [55], [56], [57].

4.2.3 Vital signs with people tracking firmware

The Vital Signs with People Tracking firmware utilizes the capabilities of the TI mmWave IWR6843ISK sensor to monitor both vital signs and the movement of individuals in an indoor environment. This firmware employs algorithms from the 3D People Counting demo for detection and tracking, allowing the localization and tracking of a single moving object in the scene. The object continues to be tracked until it leaves the scene, even if it becomes stationary by sitting or lying down. The vital sign measurements will only be accurate when the tracked person stops moving for at least 20 seconds. The radar must be placed at a height ranging from 1 to 1.5 meters, or it must be placed overhead, facing the bed. The person should be seated or lying down with the sensor pointed toward their chest, up to a distance of 5 meters [58]. The firmware processes radar data to track people and extract their heart and breath rates.

4.2.4 TI Industrial Visualizer

TI Industrial Visualizer is a software tool for visualizing and analyzing data from TI's mmWave radars. It provides an intuitive graphical interface, allowing users to easily interpret radar data in real-time. Users can visualize point cloud data, track objects, and adjust sensor configurations to optimize performance for various applications with the visualizer. The tool supports multiple radar sensors and firmware. Figure 13 is the screenshot of the visualizer that runs the 3D people counting demo firmware, which shows the detected human target in a box.

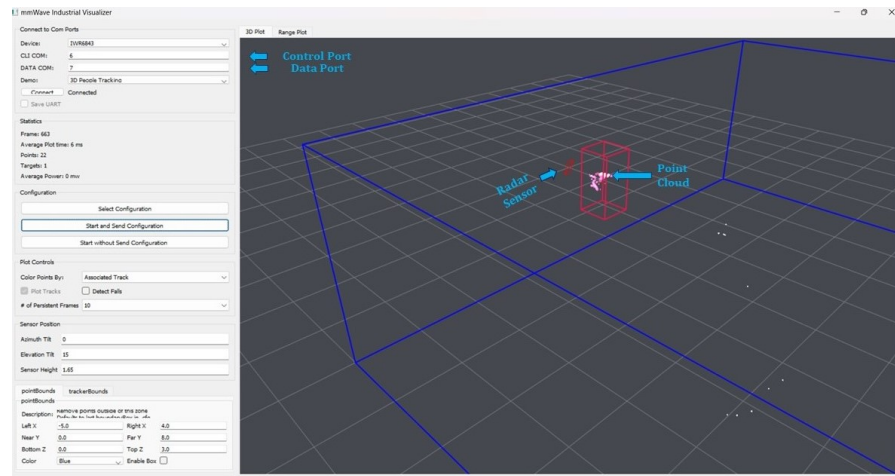


Figure 13: Screen shot of the Industrial Visualizer with 3D people counting demo firmware.

Similarly, Figure 14 shows the visualizer running the vital sign with people tracking firmware, where the detected individual's heart rate and breath rate are highlighted.

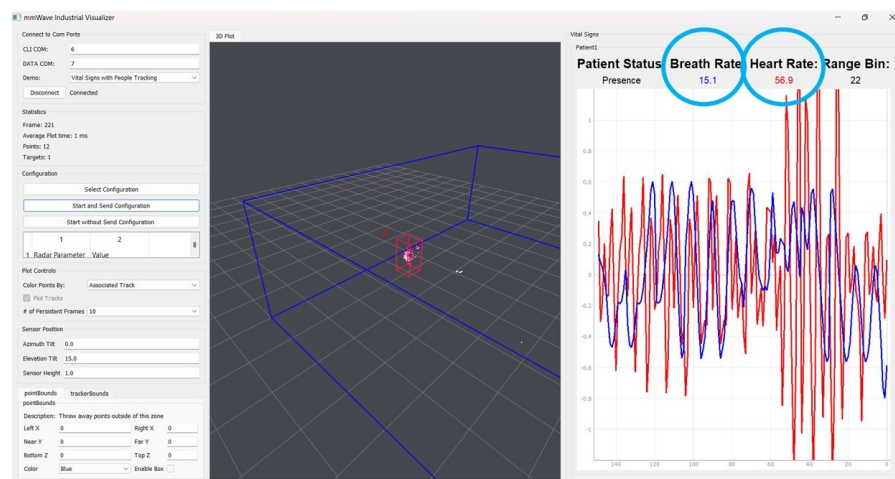


Figure 14: Screenshot of the Industrial Visualizer with Vital Signs firmware. Detected heart rate and breath rate are highlighted.

The following sections discuss the evaluation methods employed and the implementation of health monitoring systems using the IWR6843ISK radar platform and its firmware.

4.3 EVALUATION METHODOLOGY

This thesis adopts the benchmark model for sensors in smart environments, as proposed by Braun et al. [3]. As explained in the literature survey 4.3, this method involves evaluating sensor technology based on predefined features and their importance to specific applications. The evaluation methodology is structured into the following four steps, and the process is illustrated in Figure 15 :

- Feature Identification: Selects and defines the features relevant for evaluating sensor-based systems.
- Weight assignment for application: Assigns weights to each feature based on their importance to the application.
- Feature scores for the radar and radar-based systems: Identifying the scores for the features related to the radar and conducting experiments for the radar-based systems.
- Computation of the model score: Calculates the final model score for the radar-based systems by combining the feature weights and scores.

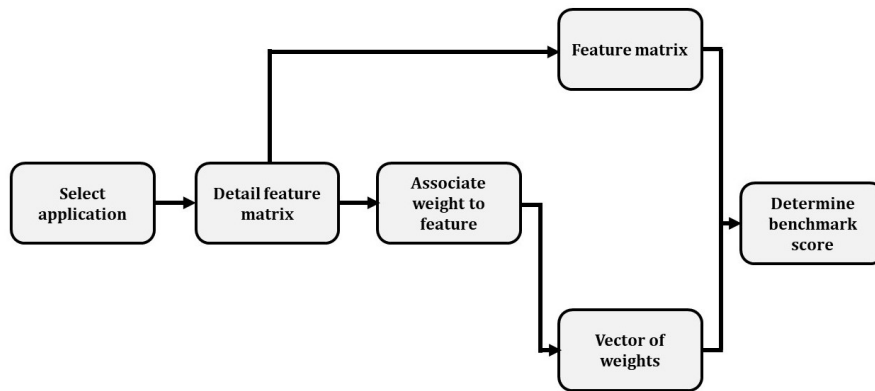


Figure 15: Process flow of the benchmark model [3].

4.3.1 Feature Identification

The first step involves identifying relevant features for the evaluation. The features that are considered in that benchmark model include resolution, update rate, detection range, unobtrusiveness, processing complexity, robustness, disturbance frequency, calibration complexity, and unique limitations [3]:

- **Resolution (res):** The level of detail a sensor can detect. Resolution refers to the smallest change in the measured physical quantity that a sensor can detect. Higher resolution means the sensor can capture finer details.
- **Update Rate (upd):** The frequency at which the sensor provides new data. The update rate indicates how often a sensor reports data within a given time period. A higher update rate means more frequent data updates.
- **Detection Range (det):** The maximum distance the sensor can detect objects. The detection range specifies the sensor's furthest distance to identify or measure a target accurately.
- **Unobtrusiveness (uob):** The degree to which the sensor remains unobtrusive in its environment. Unobtrusiveness measures how much the sensor blends into its surroundings without interfering with the environment or being noticeable. This feature is key in scenarios where the sensor should not draw attention or disturb the environment.
- **Processing Complexity (prc):** The complexity of hardware resources required for the sensor's functionality. An increase in the number of sensor nodes needed increases the complexity, and sensors that do not need further processing are the least complex.
- **Robustness (rob):** The sensor's ability to operate effectively. Robustness indicates the reliability and redundancy of the system, highlighting its ability to function despite failures. This means the sensor can withstand and continue to perform even when faced with challenging circumstances. A robust sensor is reliable and maintains functionality even during system failures.
- **Disturbance frequency (dfr):** The likelihood of the sensor being affected by external disturbances. This feature measures interference frequency from physical obstructions or other environmental noise. It ranges from very frequent to highly unlikely.
- **Calibration Complexity (cac):** The difficulty involved in calibrating the sensor. Calibration complexity involves the effort, time, and expertise required to adjust the sensor to ensure accurate measurements. It ranges from very hard to very easy.
- **Unique limitation (unl):** Unique limitation refers to the specific constraints inherent to a particular sensor. This means that some sensors may only provide binary data, such as detecting whether an object is present or not, but not offering information on how far the object is. Some other sensors, such as light or sound, are affected by ambient conditions. This feature

is graded based on how the limitation will affect a system. It ranges from very critical to not affected.

The summary of the rating criteria for the features related to sensors is shown in Table 1

Table 1: Rating criteria for features related to sensor [3].

Feature	–	-	0	+	++
Resolution	very coarse	coarse	normal	fine	very fine
Update Rate	less than once per second	slower real-time	real-time	faster real-time	more than 100 times per second
Detection Range	touch	less than one meter	less than 5 meters	less than 20 meters	more than 20 meters
Unobtrusiveness	open large system	open small system	hidden system, large exposure	hidden system, small exposure	invisible
Processing Complexity	single sensor CPU	10+ sensors CPU	single sensor embedded chip	10+ sensors embedded chip	no further processing
Robustness	single point of failure	error detection	quality of service	self-recovery	fully redundant
Disturbance Frequency	very frequent	frequent	average	unlikely	highly unlikely
Calibration Complexity	very hard	hard	normal	easy	very easy
Unique Limitations	very critical	critical	average	not critical	none

However, more than an evaluation of the sensor's features is required in a realistic home environment. The evaluation method by J. Antonio et al. [15] uses metrics such as accuracy and availability/response time for evaluating entire systems. Hence, for a better evaluation of the sensor-based systems, the system's accuracy and response time are included as extra features of the benchmark model. The rating criteria are proposed by the authors. To validate these criteria, a survey was conducted with these ratings, and participants had options to agree or propose different rating criteria. The majority of the participants agreed to the proposed criteria. The features and the finalized rating criteria are described below:

- **Accuracy (acc):** The precision of the sensor's measurements and the overall system's performance. Accuracy refers to how close the sensor's readings and the software processing outputs are to

the actual value of the measured quantity. The systems with accuracy above 95% are given the highest rating, and the systems with accuracy less than 70% are given the lowest rating. Since this accuracy feature is a new addition to the benchmark model, the divisions in ratings are finalized according to the survey conducted. These ratings are divided to evaluate systems designed for smart home monitoring healthcare applications properly.

- **Response Time (ret):** The response time is the time a system takes to process sensor data to produce a desirable output for an application. Systems with a response time of less than 10 seconds are the least preferred, and the systems with fast real-time responses, less than 100 milliseconds are the most preferred. The ratings are given one level less than the update rate in Table 1; this is because any system processes the data from the sensor, which might cause delay based on the complexity of the systems. These ratings are also finalized according to the survey results.

The summary of the rating criteria for these features is shown in Table 2

Table 2: Rating criteria for features related to sensor-based systems, according to the results of the survey conducted .

Feature	-	-	o	+	++
Accuracy	less than 70%	70 - 80 %	80 - 90 %	90 - 95 %	more than 95%
Response Time	Less than ten per second	1 s - 10 s	500ms - 1 s	100 - 500 ms	< 100 ms

In summary, 11 features are selected for evaluating the sensor-based system, consisting of the nine features for evaluating the sensor and the two additional features for evaluating the system's performance in a home environment. The next step involves assigning weights to these features in reference to the selected applications.

4.3.2 Weight assignment for application

To ensure consistency in evaluation, the author [3] proposes using a standardized five-point scale. The importance of each feature to a specific application is rated on a 5-point scale: "not important" (-), "less important" (-), "moderately important" (o), "important" (+), and "very important" (++). This rating provides a feature matrix corresponding to each application, allowing for the evaluation of a sensor-based system.

These are the five applications that are selected for evaluating the radar-based systems in the different categories of smart home health-care monitoring:

- Localization for functional monitoring,
- Human activity recognition for functional monitoring,
- Intruder detection for security monitoring,
- Fall detection for safety monitoring,
- Multiple people detection for social monitoring,
- Visitor identification for social monitoring, and
- Heart rate and breath rate detection for physiological monitoring.

The weights associated with the features based on their importance to different applications are determined, as shown in Table 3. The weights for the sensor-based features are given similar to what Braun et al. [59] have given for various applications. The weights for the system-based features are proposed by the authors. These application weights are subjective to the authors, and to address this limitation, a survey was conducted in which the participants were given the option to assign application weights. The majority of the people who took part in the survey agreed with the weights assigned by the authors.

Table 3: Resolution (res), Update Rate (upd), Detection Range (det), Unobtrusiveness (uob), Processing Complexity (prc), Robustness (rob), Disturbance frequency (dfr), Calibration Complexity (cac), Unique limitation (unl), Accuracy (acc), and Response Time (ret).

Application	res	upd	det	uob	prc	rob	dfr	cac	unl	acc	ret
Localization	o	+	++	+	-	+	o	+	-	o	+
Human activity recognition	++	+	++	+	-	+	o	+	-	+	+
Intruder detection	++	+	++	++	-	+	o	+	-	++	+
Fall detection	++	+	++	+	-	+	o	+	-	++	+
Multiple people detection	o	o	+	+	-	+	o	+	-	+	o
Visitor identification	++	+	+	+	-	+	o	+	-	+	o
Heart and breath rate detection	++	+	++	+	-	+	+	+	+	+	o

Localization: The sensor feature weights for indoor localization have been addressed by Braun et al. [59], and the reasons are discussed below. The resolution is rated average because indoor localization does not need a high resolution, as it is sufficient to know only the target’s approximate location in an indoor environment. The update rate is significant for following the trajectory of the resident’s movement. The detection range is highly important to cover large areas in an indoor environment. Unobtrusiveness is important since the sensor should not interfere with the resident or invade their privacy. Processing complexity receives a less important rating because its impact on the application is minimal, even if the sensor requires additional hardware to implement this feature. The system should be robust. The disturbance frequency is less in indoor environments compared to outdoor environments. Although there can be disturbances in a home environment due to clutter, the complexity of calibration is important to resume normal operations. The unique limitation does not affect the overall system as it is considered in the system design. Accuracy is not important, as approximate positioning is required for indoor localization [59]. Response time is important to ensure timely tracking updates.

HAR: The resolution is very important to capture fine movements and distinguish between different activities. Update rate, detection range, Unobtrusiveness, processing complexity, robustness, disturbance frequency, and calibration complexity are similar to the weights given in localization due to the characteristics of the indoor environment. Accuracy is important to classify the activities correctly. Response time is important for the timely classification of different activities.

Intruder Detection: This application greatly emphasizes resolution, as detailed imagery is necessary to identify intruders accurately. The update rate is rated as important to ensure timely detection updates. The detection range is very important to cover the maximum area inside the indoor environment. Unobtrusiveness is rated very important, as sensors must remain hidden from intruders. Processing complexity, robustness, disturbance frequency, and calibration complexity are similar to weights given in localization due to the characteristics of the indoor environment. Accuracy is very important to detect intruders correctly to ensure security for the residents in the home environment. Response time receives an important rating to ensure timely security alerts.

Fall Detection: For fall detection, resolution and detection range are rated as very important, as detailed imagery and the ability to detect falls over a range are crucial for this application. The update rate receives an important rating to ensure timely detection. Unobtrusiveness, processing complexity, robustness, disturbance frequency, and calibration complexity are similar to the weights given in localization due to the characteristics of the indoor environment. Accuracy is rated as very important, as precise detection of falls is crucial for the resident's safety. Response time receives an important rating to ensure timely alerts.

Multiple People Detection: This application's resolution is moderately important because only the presence of multiple people needs to be detected. The detection range is given a lower rating than other applications since the presence of visitors is in a smaller area in the home environment. The update rate is rated as moderately important, as frequent updates are unnecessary for multiple people detection. Unobtrusiveness is important to ensure the sensor does not interfere with activities. Processing complexity, robustness, disturbance frequency, and calibration complexity are similar to weights given in localization due to the characteristics of the indoor environment. Accuracy receives an important rating to identify multiple people in the environment properly. Response time is rated moderately important, as immediate updates are less critical for detecting multiple people.

Visitor identification: The resolution for identifying visitors is very important to capture the unique features of each visitor. The detection range is given a lower rating than other applications since the presence of visitors is in a smaller area in the home environment. The

update rate is important to capture the visitor's movements and identify them properly. Unobtrusiveness is important to ensure the sensor does not interfere with activities. Processing complexity, robustness, disturbance frequency, and calibration complexity are similar to weights given in localization due to the characteristics of the indoor environment. Accuracy receives an important rating when correctly identifying visitors. Response time is rated moderately important, as immediate updates are not required.

Heart and breath rate detection: For heartbeat and breath rate detection applications, resolution is rated as very important due to the need for precise measurements. The update rate receives an important rating to ensure timely monitoring updates. The detection range is very important for continuously monitoring the parameters in the entire environment. Unobtrusiveness is important to ensure the sensor does not interfere with activities. Processing, robustness, and calibration are rated the same as localization. The disturbance frequency is rated above localization as disturbances caused might affect the measurement of these parameters. Accuracy is given an important rating as accurate readings are necessary for monitoring older adults. Response time is given an average rating because real-time updates are not necessary in a home environment.

4.3.3 *Feature scores for the radar and radar-based systems*

This section discusses how the feature scores are assigned to the radar. This part is divided into two categories: the first describes the scoring of the nine features related to the sensor, and the second is the scoring of the additional features associated with the whole system. The rating criteria for these features are shown in Table 1. These features can be scored using the datasheet of IWR6843ISK and also based on how the author scored for other sensors in [59].

The resolution of the radar is very coarse, as it can even detect the slightest movements, like breathing and heart rates. The samples of each data coming from the radar are in fast real-time. The detection range of the sensor is around 10 m. The radar platform is reasonably unobtrusive, as it has only a small exposure and does not interfere with the resident's daily living activities. Noise signals and disturbances are sometimes generated due to reflections from the clutter in the home environment. IWR6843ISK is a single-sensor embedded chip. Sensors, including cameras, capacity proximity sensors, ultrasound, and radio, are given an average score for robustness [59]. Therefore, radar can also be rated as average for robustness. The calibration complexity of radar is neither easy nor difficult. Radar signals are not affected by ambient conditions like light or sound, but occlusion can be considered a limitation. This can sometimes affect

the system. Therefore, the ratings for the features of IWR6843ISK are shown in Table 4.

Table 4: Resolution (res), Update Rate (upd), Detection Range (det), Unobtrusiveness (uob), Processing Complexity (prc), Robustness (rob), Disturbance frequency (dfr), Calibration Complexity (cac), and Unique limitation (unl).

Sensor	res	upd	det	uob	prc	rob	dfr	cac	unl
IWR6843ISK radar sensor	++	+	+	+	o	o	o	o	o

The sensor-based features are evaluated based on the experiments conducted in HINT for each application. These are mentioned in the following sections related to each type of monitoring system.

4.3.4 Computation of model score

The proposed model determines a benchmark score for the application and sensor-based system that performs this application. As previously explained, for each application, a detailed feature matrix, along with a vector of associated weights that are applied to the sensor-based systems that perform this application, is considered to calculate the benchmark score.

Different ratings are assigned numeric values: 0.00 (-), 0.25 (-), 0.5 (o), 0.75 (+), and 1.00 (++) for the mathematical calculations.

For each application a_j and feature f_k , a weight w_{f_k, a_j} is defined, representing the importance of feature f_k for application a_j . Each sensor technology s_l is evaluated against each feature f_k , resulting in a feature score r_{s_l, f_k} .

The overall benchmark score b_{s_l, a_j} for sensor s_l in application a_j is calculated as [3]:

$$b_{s_l, a_j} = \sum_{k=1}^o w_{f_k, a_j} \cdot r_{s_l, f_k} \quad (6)$$

Additionally, a normalized benchmark score $b_{s_l, a_j, norm}$ is derived to facilitate comparison across different applications [3]:

$$b_{s_l, a_j, norm} = \frac{b_{s_l, a_j}}{\sum_{k=1}^o w_{f_k, a_j}} \quad (7)$$

In these equations:

- w_{f_k, a_j} is the weight assigned to feature f_k for application a_j . This weight represents the importance or relevance of a particular feature in the context of a specific application, as determined by the authors based on domain knowledge and requirements.

- r_{s_l, f_k} is the feature score for sensor s_l with respect to feature f_k . This score indicates how well the sensor performs regarding a specific feature, as assessed by the authors using performance metrics or empirical data.
- b_{s_l, a_j} is the overall benchmark score for sensor s_l in application a_j . It is calculated by summing the weighted feature scores, providing a comprehensive measure of the sensor's suitability for the application.
- $b_{s_l, a_j, \text{norm}}$ is the normalized benchmark score obtained by dividing the overall benchmark score by the sum of the weights for the application. Normalization ensures the scores are comparable across different applications by adjusting for varying weight distributions.

Normalization is essential to ensure fair comparisons across applications with weight distributions of different importance [3]. It adjusts the benchmark scores to a standard scale, allowing for a more balanced and meaningful comparison of sensor performance across diverse applications. Without normalization, applications with higher total weights could disproportionately influence the benchmark scores, leading to biased conclusions.

Additionally, the author identified central tendency bias in the article, a common phenomenon where ratings cluster around the middle of a scale rather than evenly distributed across its entire range. This effect is commonly found in Likert-scale questionnaires and rating systems [60]. The benchmark scores in the study exhibited this bias, predominantly centering around 0.5 on a normalized scale. This clustering is particularly significant, as the differences between scores are less pronounced when they are close to this average [3]. To address this issue and enhance the readability and interpretability of the data, the authors applied the following adjustment equation:

$$m_{b, \text{norm}} = (b_{s, \text{norm}} + 0.5)^{\alpha} \quad (8)$$

They experimented with different values for the parameter α , explicitly using 1, 5, and 10. These adjustments aimed to spread the scores more evenly across the scale, thereby reducing the impact of central tendency bias and making the differences between benchmark scores more distinguishable.

4.4 MMWAVE SENSOR-BASED SMART HOME HEALTH MONITORING APPLICATIONS

The five distinct monitoring categories implemented for the evaluation of mmWave radar-based smart home monitoring systems in

healthcare for older adults living alone are functional (localization and human activity recognition), safety (fall detection), security (intruder detection), social (detecting multiple people detection and visitor identification), and physiological (heart and breath rate detection).

4.4.1 *Functional monitoring*

This section discusses the localization and human activity recognition systems implemented in functional monitoring. The experiments conducted for the evaluation of these systems are also discussed.

4.4.1.1 *Localization*

The localization system gives out the coordinates of the resident's position inside the HINT. The first step is to install the radar platform in the HINT. A tripod setup and a 3d printed base are used to secure the radar in place, offering the flexibility to adjust the radar's height and azimuth tilt as needed, as shown in Figure 16.



Figure 16: Radar's physical setup.

The radar was placed in the center of the HINT, as shown in Figure 17, as it does not interfere with the resident's daily living activities and gives maximum coverage of the environment. The height of the radar is 1.65 meters, and the elevation angle is 15 °.

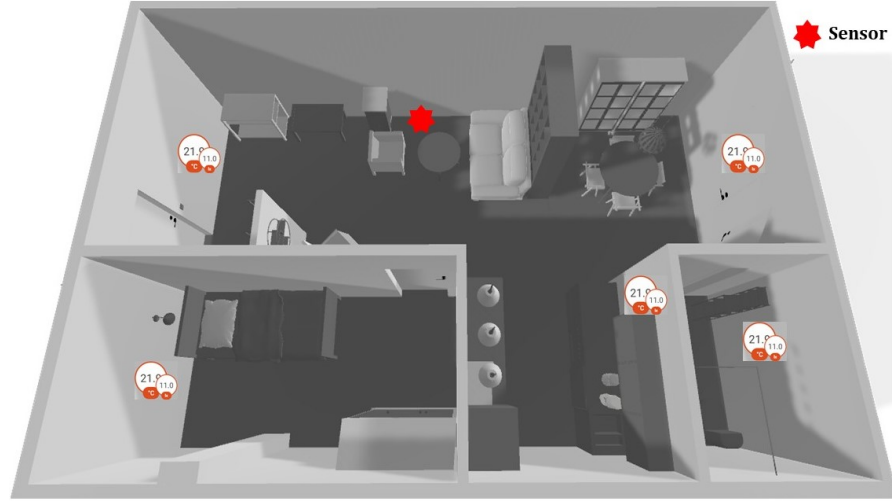


Figure 17: Radar's position in HINT.

The track data from the 3D people counting demo firmware gives the x and y coordinates of the person detected. This data is used for the localization.

To compute the accuracy of localization, two reference points are considered in each of the five distinct areas of HINT. These points fall within the radar's coverage area, as shown in Figure 18. The x and y coordinates of these points in meters with reference to the sensor as its origin is also highlighted in the figure.

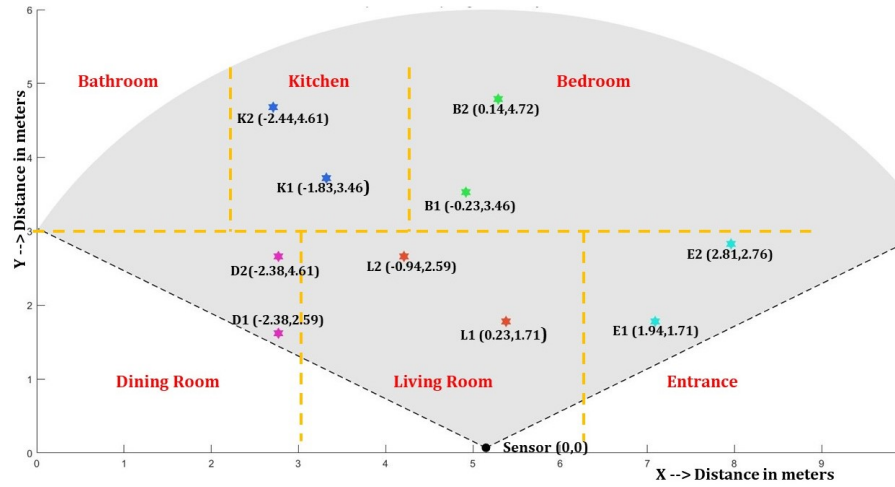


Figure 18: Reference points for computing the accuracy of the localization.

The x and y coordinates of the track data from the sensor platform are stored for the person standing in these reference points. The average of 10 readings from a single point is taken as the measured point. Euclidean distances of the measured points and reference points are calculated. The method followed in [15] to rank the localization accuracy of the competitor's system is based on the percentile of the

Euclidean distance. In this thesis, the system's accuracy is given a rating to be included in the benchmark model.

The rating is based on the Euclidean distance; from recent works, as mentioned in section 3.1, the error distance achieved by feasible indoor localization systems are 0.64m and 0.42 m. So, this thesis gives the highest rating for the localization system for error distances lesser than 25 cm, which is less than the recent works. The lowest rating is given for errors greater than one meter. The rating criteria are shown in Table 5. Additionally, this rating system was validated through a survey, where 58 out of 59 participants agreed with the proposed ratings.

Table 5: System accuracy rating based on Euclidean distance.

Rating	-	-	0	+	++
Euclidean distance	>1 m	1 m to 75 cm	75 cm to 50 cm	50 cm to 25 cm	<25 cm

4.4.1.2 Human Activity Recognition

Basic human activities like standing, sitting, and walking are included in the HAR system. A basic deep learning model is implemented, which classifies these three activities. More than 15 minutes of the resident walking in the HINT is recorded and labeled as walking. Five minutes of the resident sitting in three different seats are recorded and labeled as sitting. Similarly, five minutes of the resident standing in three different positions are recorded and labeled as standing. Approximately 45 minutes of data is taken to train the HAR model.

The collected radar point cloud data includes X, Y, and Z coordinates, Doppler velocity, SNR, and the quality index. The data from the radar sensor module is saved into a comma-separated values (CSV) file. From there, it is further processed to reject outlier points and to standardize the number of points in each frame. Finally, the data is converted to Python tensor format, as shown in Figure 19. This conversion ensures that the data is in a suitable format for inputting into the deep learning model.

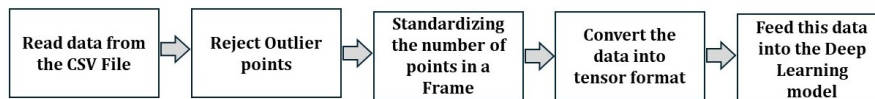


Figure 19: Data processing steps.

In an indoor environment, numerous obstacles are present along the path, resulting in scattered points outside the central cluster. There-

fore, outlier points are present in the data captured by the radar sensor. Figure 20 illustrates the presence of these outlier points.

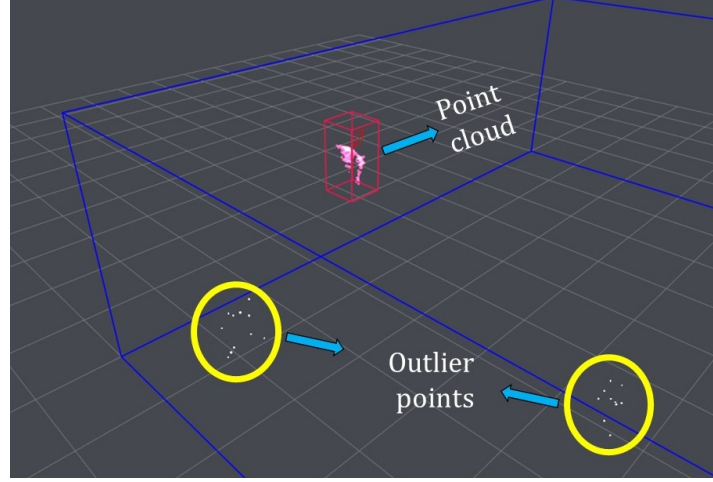


Figure 20: Outlier points displayed on TI Industrial Visualizer.

Data processing is essential to standardize each data frame and eliminate outlier points within a frame. The outliers are mostly present away from the actual range of the detected person, and the person can be standing at different angles to the radar. Hence, filtering is done in the x and y axes. The maximum number of points detected are from the torso region of the person [55]. Hence, the median is located near the middle of the torso. According to the data set released by M. Hitka et al., [61], the average length of arms spread while standing is approximately 1.8 meters. So, taking half of that length (0.9 m) gives an approximate distance from the middle of the torso to an arm, as shown in Figure 21.

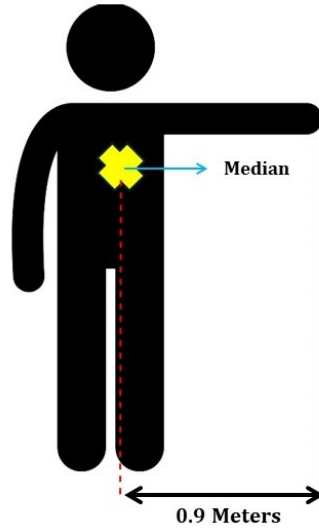


Figure 21: Threshold from the median in X and Y axes.

To understand a typical frame, a single frame's x and y points are plotted as shown in Figure 22. The median for the x and y axes are calculated. A threshold of ± 0.9 meters is added from the median on both axes and is shown in the plot as blue lines. The points outside this boundary are removed as these are outliers.

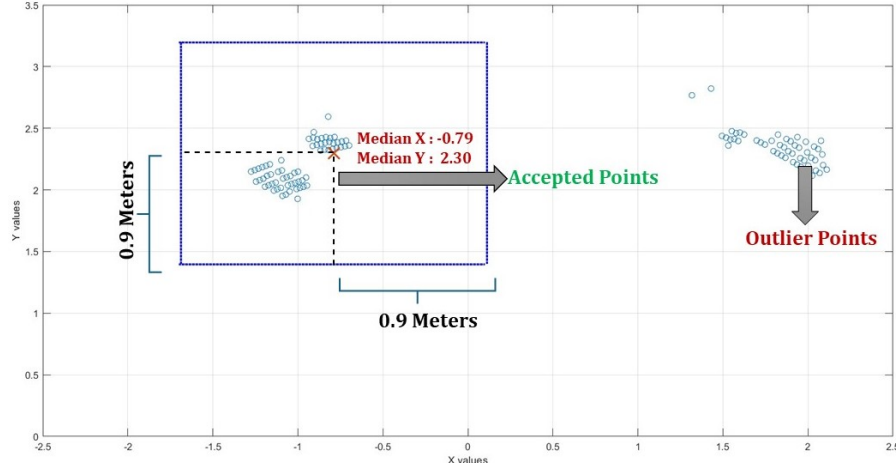


Figure 22: Scatter plot of the number of points in a frame, showing the median and the outlier points.

After rejecting the outliers, the next step is ensuring all frames have equal points. The number of points in a single frame was analyzed by writing the point cloud TLV to a Comma-Separated Values (CSV) file. The number of points in a single frame while walking was the highest, and the number of points in a single frame while sitting was the lowest. The frame size varied from 10 to more than 100 while walking, with the most frequent frame size being in the range of 20 to 60. The frame sizes for sitting were less than 40. Hence, the fixed frame size is considered to be 40. If the points are less than 40, random copies of the existing points are made. If the points exceed 40, the 40 points with the highest SNR values are retained, and the rest are removed. SNR and the quality index did not yield good test results in the experiments conducted in HINT, so these are removed as they do not aid in learning the basic activity. Ten frames are appended to distinguish the activities, and they are converted into tensor format and split into training, validation, and testing datasets. For training, an NVIDIA RTX A5000 GPU is utilized. Learning both temporal and spatial features is necessary for recognizing human activity. Therefore, the deep learning model chosen is CNN + LSTM. The CNN helps capture spatial features, while the LSTM component captures temporal features. A fully connected layer is added as the last layer for classification. Cross-entropy is the loss function, with the Adam optimizer and a scheduled learning rate. Figure 23 illustrates the layers of the proposed model.

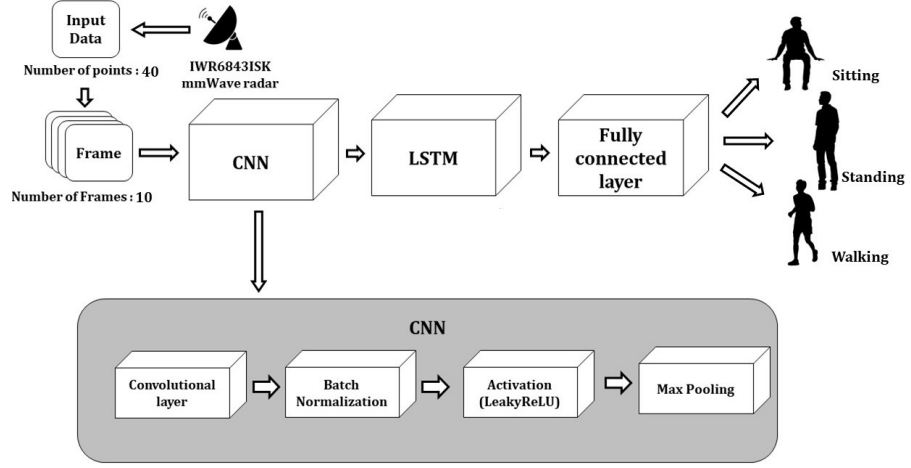


Figure 23: CNN with LSTM for HAR.

The model was trained for 100 epochs, as the training loss did not improve after this point. Hyper-parameters were tuned to enhance the model's accuracy. The accuracy is calculated on the test dataset to assess the HAR system's performance using Equation 9.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (9)$$

Where:

- **TP (True Positives):** The number of correctly identified positive cases.
- **FP (False Positives):** The number of negative cases incorrectly identified as positive.
- **FN (False Negatives):** The number of positive cases incorrectly identified as negative.
- **TN (True Negatives):** The number of correctly identified negative cases.

Additionally, the model was deployed in real-time within the HINT environment to evaluate its response time in predicting the basic activities of the target.

4.4.2 Safety Monitoring

This section discusses the system implemented as part of the safety monitoring and the experiments conducted to evaluate the system. The fall detection algorithm is implemented as part of the Safety monitoring system. As mentioned the resident's height is also received from the sensor platform. This height is continuously monitored to detect falls in the environment.

To cover the maximum area for detecting falls, the radar is placed in the same position mentioned in section 4.4.1.1. Based on this position, the number of frames and the percentage drop in height needed to identify a fall must be calibrated.

Two experiments are conducted in the environment for the calibration; the author walks around the environment, sits on a chair, and gets up in the first one. In the second one, the author walks and simulates a fall. The height is recorded in both experiments and plotted with the number of frames as shown in Figure 24. The height drops to around 75 % when the person sits, and it takes more than 20 frames. Whereas while falling, the height drops to less than 60% in 20 frames. So, the number of frames to continuously monitor the drop in height is set as 20, and the percentage threshold to detect a fall is set as 60%.

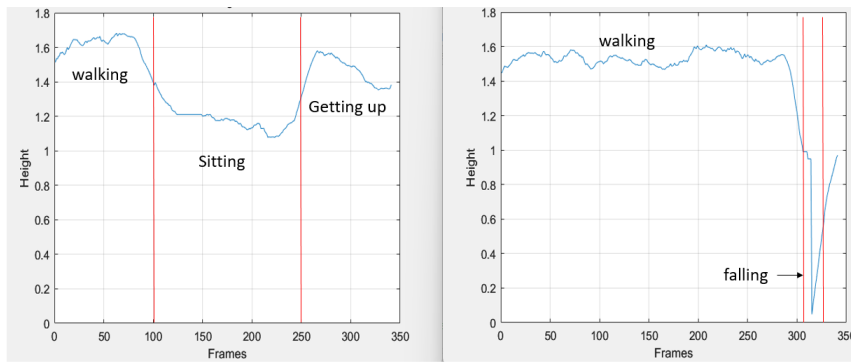


Figure 24: Height plots for walking and sitting (left), walking and falling (right).

The initial state of the system is named the tracking state. If the height of the detected target drops to less than 50% in 10 frames, then the system goes to the fall detected state. However, in a real scenario, if an older adult falls seriously, they cannot get up immediately. Considering that, an extra state, called fall confirmed, is added. The height at that point is saved as fall height, and a timer is started. In the fall-detected state, the height is continuously monitored till the defined time. If the current height is greater than the fall height + 10% of the fall height, the fall is not confirmed, and the system returns to the tracking state. If the system stays in the state of fall detected for the defined time, the fall is confirmed.

Sometimes, the resident falls behind furniture in the living room or falls out of sight of the radar. In these cases, the radar loses sight of the resident, and the track information received from the sensor is lost. Due to the absence of track information, height cannot be monitored. To consider such scenarios, a fall is confirmed if the drop in height is detected and the sensor loses sight of the target for the defined time. If the resident gets up, the sensor receives the tracking information again.



Figure 25: Camera view of the kitchen.

In the kitchen area, when a person is falling, the drop in height might not always be detected as the distance to the sensor is maximum in the area and because of the 30° elevation angle in the field of view of the radar. Figure 25 is the photo taken by a camera from the sensor's position. The sensor can lose track of the person in the kitchen area only if the person is lying down, as seen in the photo. So, to detect all falls in the kitchen area, a fall is confirmed if the radar loses track of the person in that area for the defined time. The state diagram of the algorithm is shown in Figure 26

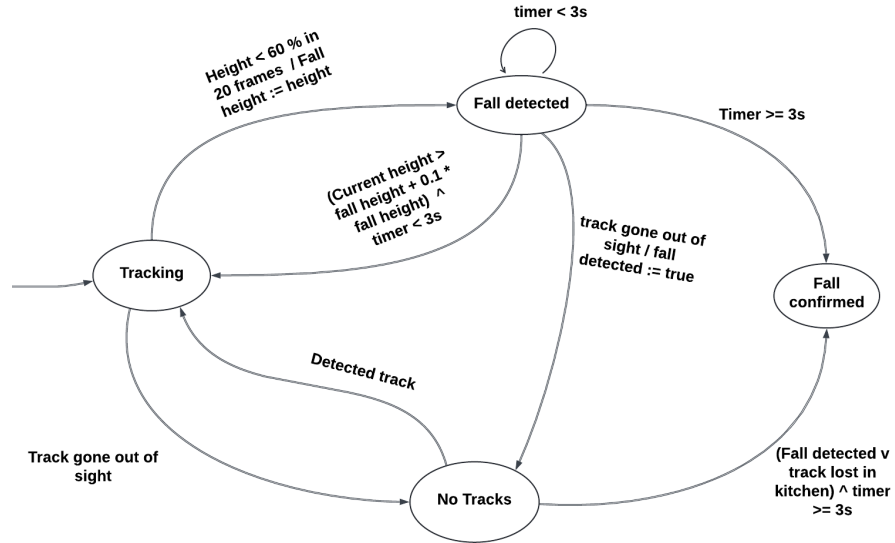


Figure 26: State diagram of the fall detection algorithm.

As previously discussed in section 3.2, there are cases where older adults take longer time to get up and some cases where they stay lying on the floor [39], [38]. Therefore, the defined time is set as three seconds to balance timely detection with minimizing false alarms.

To evaluate the accuracy of the fall detection algorithm, the resident simulates falls of three kinds: forward, backward, and sideward. The entrance and dining room areas are not considered for accuracy computation as they fall outside the border of the radar's field of view. The three kinds of falls are simulated in the living room, kitchen, and bedroom within the radar's field of view. The fall detection system's output is checked five times in each area for the three falls. Other actions like sitting and bending down are performed multiple times to check for false positive cases. The system's output is recorded and used for computing the accuracy of the system. The response time of the algorithm is 3 seconds as the algorithm takes 3 seconds to confirm a fall.

4.4.3 *Security Monitoring*

This section discusses the system implemented as part of the security monitoring and the experiments conducted for the evaluation of the system. Intruder detection is implemented as part of the security monitoring system. This intruder detection system is activated manually when the resident leaves the smart home. After the system is activated, if a human presence is detected, the system identifies the presence as an intruder. The track information from the firmware is used to detect human presence. The position of the radar system is placed at the same location mentioned in section 4.4.1. At this position, the radar is in the middle of both the entrance area and the emergency exit area. If the intruder attempts to enter through any doors, the system can detect the intruder properly.

Participants are instructed to enter the HINT as intruders from the two entry points. Different intrusions, including normal walking, crawling, and crouching, are simulated to represent how an intruder might approach. All the activities are done from both the entrances to all the different areas in the HINT. When the system is activated in security mode, the output of the system is recorded while performing these intrusions to compute the accuracy.

Experiments are conducted to identify false positives, including tests to determine whether the system mistakenly identifies nonhuman movements, such as those of a vacuum cleaner bot, as intrusions.

4.4.4 *Social Monitoring*

This section discusses the multiple people detection and visitor identification systems implemented in social monitoring. The experiments conducted for the evaluation of these systems are also discussed.

4.4.4.1 *Multiple people detection*

The social life of the person living alone in a smart home can be monitored by detecting multiple people. The firmware of the radar sensor supports the detection and tracking of multiple people. Thus, this firmware can be used to detect visitors in the smart home.

The position of the radar is kept in the same position (Figure 17) to get maximum coverage. One person is considered the resident, and three are considered visitors. The resident is always present in the HINT, and each visitor is instructed to enter individually. The number of times the system can detect the visitors is taken to compute the system's accuracy. All combinations of each visitor and resident situated in the same area and different areas are considered. Experiments are done to study the effects of occlusion. The resident and the visitors are asked to stand in the same line to check if the radar can detect each person separately. These experiments are also done in all the areas that fall in the field of radar vision.

Experiments are also done to study if false positive cases are there. Multiple people are detected or not if the resident is alone and opens or closes the door to the bedroom, shelves, or refrigerator door. Similarly, when the resident is alone and walking in the home, checks are conducted for the presence of any ghost targets. Also, similar to the false positive checks done in 4.4.3, the system checks if vacuum cleaner bot movements are mistakenly detected as visitors when the resident is present inside the HINT.

4.4.4.2 *Visitor Identification*

In addition to tracking the in-person social activity by detecting multiple people in the home environment, recognizing social visitors and keeping track of the visits can enhance social monitoring of older adults living alone [46]. Visitors can be classified using the radar data based on their gait patterns. A basic deep learning model is implemented to learn these gait features and classify two different individuals. One hundred twenty minutes of data are recorded for both individuals while walking and performing different activities in HINT to capture their gait. Hence, a total of 240 minutes of data is used to get the data set required for gait recognition. The recorded radar point cloud data is processed in the same way as mentioned in 4.4.1.2. It was found that each frame has varying sizes of points; some frames have more than 100 points, and some frames have fewer than 10 points. This variance is due to different activities. The points detected are fewer when standing or sitting idle than when the person is moving. The minimum number of points detected in a frame was 5, and the maximum number detected in a frame was 120. The most frequent frame sizes were approximately in the range of 60-80. Therefore, 60, 70, and 80 points were taken in a single frame to standardize

the number of points. The test accuracy for 70 points gave the best results, so 70 points per frame were chosen. If the points are fewer than 70, random copies of the existing points are made. If the points exceed 70, the 70 points with the highest SNR values are retained, and the rest are removed. Similar to HAR, the SNR and quality index were removed as they did not aid in learning gait features. To properly learn the gait pattern of each individual, more frames are needed than 10, which is used in HAR. Fifty frames were appended and then converted into the tensor format and split into the train, validation, and test datasets. For training, an NVIDIA RTX A5000 GPU is utilized. Learning both temporal and spatial features is necessary for recognizing gait. Similar to HAR, a model using CNN combined with LSTM is also used in this application. Figure 27 illustrates the layers in the model.

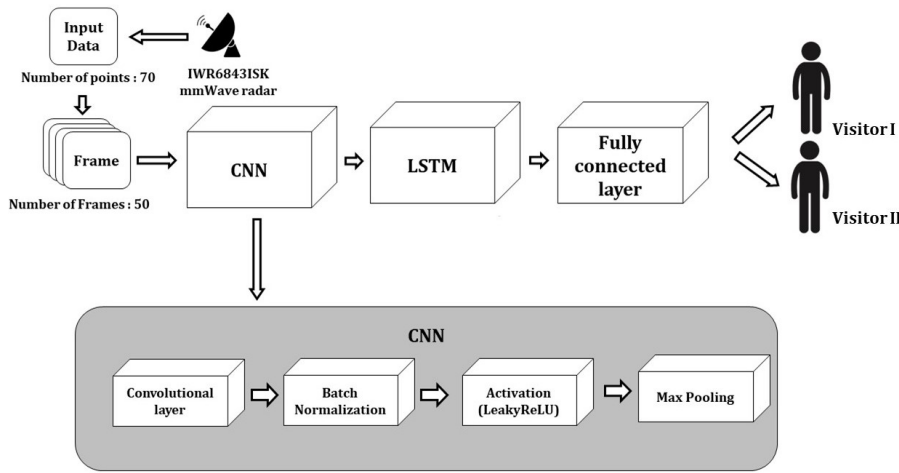


Figure 27: CNN with LSTM for visitor identification.

Two hundred epochs were utilized for training, as the training and validation losses did not decrease after that. The hyper-parameters are tuned as needed to train the model for better accuracy. The accuracy of the model on the test data set is computed to evaluate the prediction of the HAR system. The model is run in real-time in HINT to check the response time of the deep learning model for the prediction of the basic activities.

4.4.5 Physiological Monitoring

This section discusses the experiments conducted to evaluate the radar platform's heart and breath rate detection. The firmware used for the heart and breath rate is Vital signs and people tracking. The radar's height is changed to 1 meter, with the same elevation tilt of 15 degrees from the physical setup mentioned in the 4.4.1.1. As discussed in Section 4.2.3, to detect both vital signs, the resident has to sit facing the

radar pointed toward their chest, and the person has to stay stationary for 20 seconds for the system to give stable readings. It is important to note that the person's breathing should be deep enough to induce noticeable chest movements. If there is minimal or no chest movement, the radar may not detect vital signs effectively. The resident is instructed to sit on the chair situated 1.5 meters from the sensor. The TI Industrial Visualizer tool is used to visualize both readings. These readings are compared with the existing sensing device used in HINT, the Onyx3 Oximeter. It complies with ISO 10993-1, biological evaluation of medical devices [62]. Ten readings of both breath and heart rate from the radar and the oximeter are taken for comparison. The oximeter readings are considered the actual value, and the readings from the radar are taken as the measured value.

In [49], the authors define the accuracy of vital sign detection, stating that an estimation error of less than 2.5 bpm is considered reliable. Based on this criterion, a rating system is proposed in Table 6. Additionally, this rating system was validated through a survey, where all the 59 participants agreed with the proposed ratings.

Table 6: System accuracy rating based on estimation error.

Rating	–	-	0	+	++
Estimation error	>20 bpm	10 to 20 bpm	5 to 10 bpm	2.5 cm to 5 bpm	<2.5 bpm

The results of the experiments conducted and the corresponding discussions on the results of the distinct applications are described in the following chapters.

RESULTS

This chapter presents the evaluation results of all the applications in different monitoring systems. The benchmark score is then computed based on the results of each application.

5.1 FUNCTIONAL MONITORING

This section presents the results associated with both localization and human activity recognition.

5.1.1 Localization

The radar placement was done manually by looking into the environment's layout for the optimal position that gives maximum coverage. The scatter plot of the X and Y coordinates from the track data in this position is plotted as shown in Figure 28 to show the coverage of the current position of the radar.

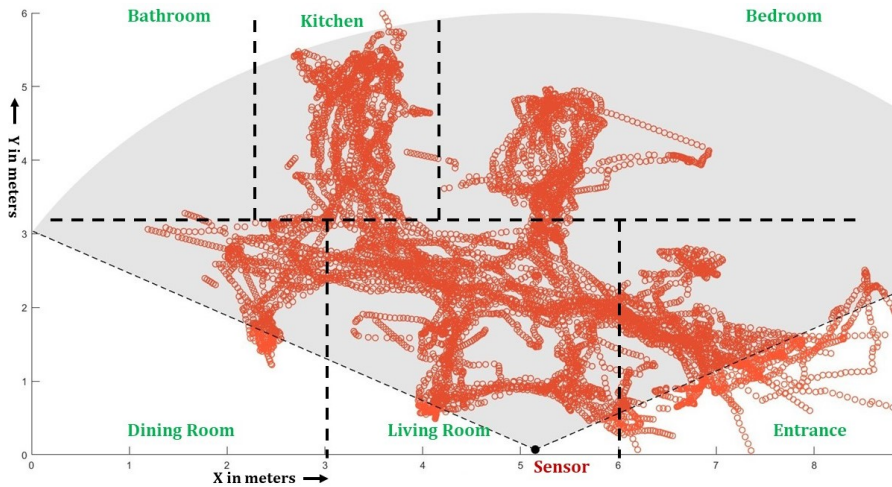


Figure 28: Scatter plot of the radar's coverage inside HINT.

Since the same firmware is used for all the other applications, the same position and configuration are used in those as well. The feature ratings for the localization system are based on the criteria outlined in Table 2 in Section 4.3.1 and are discussed as follows:

- **Accuracy:** The mean absolute error, which is the Euclidean distance of the reference point and measured point in each area, is shown in Table 7.

Table 7: Localization accuracy in distinct areas.

Area	Mean absolute error (Euclidean distance) in cm
Entrance	59.99
Living room	18.18
Bedroom	17.12
Kitchen	16.87
Dining room	50.70

Table 7 shows that the entrance and dining room have higher errors when compared to the other areas of HINT. Since these areas are at the border of the radar sensor’s coverage, and the objective is to evaluate the radar module’s capability to localize the target, the entrance and dining room’s localization points are not considered for computing the system’s accuracy. The average Euclidean distance in the living room, kitchen, and bedroom is 17.39 cm, which is less than 25 cm. As per the rating criteria mentioned in Table 5 in Section 4.4.1.1, the accuracy of the system is rated as (++) for this application.

- **Response Time:** The response time of the localization system was the same as the update rate of the radar platform, as the firmware’s output was directly used for localization. Hence, a (++) rating is assigned for response time, as the platform runs in real-time.

The benchmark score for localization is calculated based on the application ratings and feature scores for the radars from Table 3 and Table 4 in Section 4.3.2 and Section 4.3.3 respectively. Table 8 shows the combined numerical weights of the application and the feature score for the sensor and localization system.

Table 8: Resolution (res), Update Rate (upd), Detection Range (det), Unobtrusiveness (uob), Processing Complexity (prc), Robustness (rob), Disturbance frequency (dfr), Calibration Complexity (cac), Unique limitation (unl), accuracy (acc), and response time (ret).

Features	res	upd	det	uob	prc	rob	dfr	cac	unl	acc	ret
Application Weight	0.5	0.75	1	0.75	0.25	0.75	0.5	0.75	0.25	0.5	0.75
Feature score	1	0.75	0.75	0.75	0.5	0.5	0.5	0.5	0.5	0.75	1

The equations 6 and 7 in Section 4.3.4 are used to compute the benchmark score (b_{s_l, a_j}) and normalized benchmark score ($b_{s_l, a_j, \text{norm}}$) for localization application from the values in Table 8.

$$b_{s_l, a_j} = 5$$

$$b_{s_l, a_j, \text{norm}} = 0.7142$$

5.1.2 Human activity recognition

The feature ratings for the HAR system are based on the criteria outlined in Table 2 in Section 4.3.1 and are discussed as follows:

- **Accuracy:** The confusion matrix of the HAR model on unseen test data is shown in Figure 29

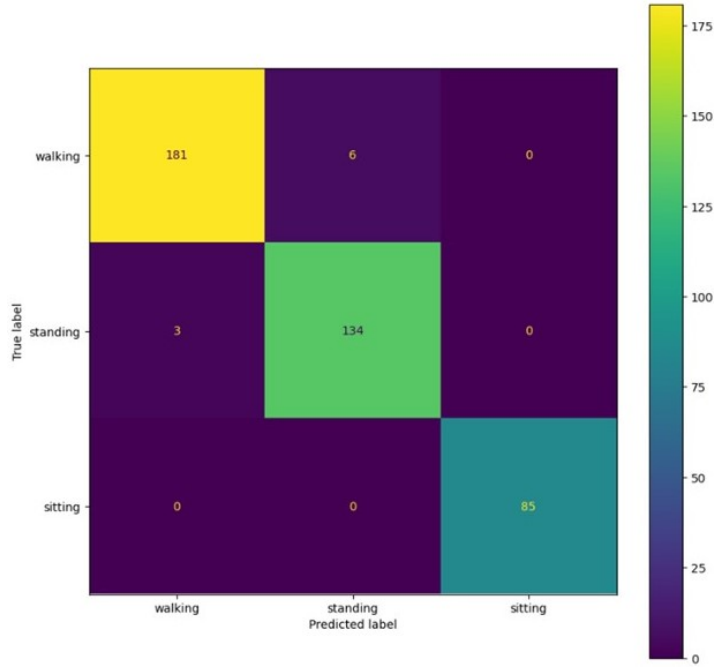


Figure 29: Confusion matrix of HAR on the test data set.

The performance of the HAR model was evaluated by computing the accuracy from Equation 9 in Section 4.4.1.2. The accuracy for the independent activities of walking, standing, and sitting are 96.84%, 97.81%, and 100%, respectively. The overall accuracy achieved by the model was 97.80%. Hence, the accuracy of the system is given as a (++) rating.

- **Response time:** The system's average response time to predict an activity was 0.33 seconds. Therefore, the rating is given as (+).

The benchmark score for HAR is calculated based on the application ratings and feature scores for the radars from Table 3 and Table 4 in Section 4.3.2 and Section 4.3.3 respectively. Table 9 shows the combined numerical weights of the application and the feature score for the sensor and HAR system.

Table 9: Resolution (res), Update Rate (upd), Detection Range (det), Unobtrusiveness (uob), Processing Complexity (prc), Robustness (rob), Disturbance frequency (dfr), Calibration Complexity (cac), and Unique limitation (unl)accuracy (acc), and response time (ret).

Features	res	upd	det	uob	prc	rob	dfr	cac	unl	acc	ret
Application Weight	1	0.75	1	0.75	0.25	0.75	0.5	0.75	0.25	0.75	0.75
Feature score	1	0.75	0.75	0.75	0.5	0.5	0.5	0.5	0.5	1	0.75

The equations 6 and 7 in Section 4.3.4 are used to compute the benchmark score (b_{s_l, α_j}) and normalized benchmark score ($b_{s_l, \alpha_j, \text{norm}}$) for HAR application from the values in Table 9

$$b_{s_l, \alpha_j} = 5.4375$$

$$b_{s_l, \alpha_j, \text{norm}} = 0.7250$$

5.2 SAFETY MONITORING

This section shows the results and computation of the benchmark score of the fall detection application. The feature ratings for the fall detection system are based on the criteria outlined in Table 2 in Section 4.3.1 and are discussed as follows:

- **Accuracy:** The results of the application on the simulated falls conducted in HINT are shown in Table 10

Table 10: Detection of falls in different areas and actions.

Area	Action	Falls detected	Not detected
Living room	Fall	14 ^{TP}	1 ^{FN}
Bedroom	Fall	12 ^{TP}	3 ^{FN}
Kitchen	Fall	13 ^{TP}	2 ^{FN}
Living room	Non-fall	0 ^{FP}	15 ^{TN}
Bedroom	Non-fall	0 ^{FP}	15 ^{TN}
Kitchen	Non-fall	3 ^{FP}	12 ^{TN}

As the dining and entrance areas fall on the border of the radar's field of view, the application's outputs in these areas were not considered in the accuracy computation. From the table, the parameters required to compute the accuracy are as follows:

- **True Positives (TP):** Falls correctly detected as falls (39).
- **False Positives (FP):** non-falls detected as falls (3).
- **False Negatives (FN):** falls not detected as falls (6).
- **True Negatives (TN):** non-falls correctly detected as non-falls (42).

The accuracy of the system calculated using Equation 9 in Section 4.4.1.2 is 90.00 %, so the rating for the accuracy of the fall detection system is given as (+).

- **Response time:** Since the response time of the system is approximately 3 seconds, the rating is given as (-).

The benchmark score for fall detection is calculated based on the application ratings and feature scores for the radars from Table 3 and Table 4 in Section 4.3.2 and Section 4.3.3 respectively. Table 11 shows the combined numerical weights of the application and the feature score for the sensor and fall detection system.

The equations 6 and 7 in Section 4.3.4 are used to compute the benchmark score (b_{s_l, a_j}) and normalized benchmark score ($b_{s_l, a_j, \text{norm}}$) for fall detection application from the values in Table 11

$$b_{s_l, a_j} = 5.0625$$

$$b_{s_l, a_j, \text{norm}} = 0.6532$$

Table 11: Resolution (res), Update Rate (upd), Detection Range (det), Unobtrusiveness (uob), Processing Complexity (prc), Robustness (rob), Disturbance frequency (dfr), Calibration Complexity (cac), and Unique limitation (unl), accuracy (acc), and response time (ret).

Features	res	upd	det	uob	prc	rob	dfr	cac	unl	acc	ret
Application Weight	1	0.75	1	0.75	0.25	0.75	0.5	0.75	0.25	1	0.75
Feature score	1	0.75	0.75	0.75	0.5	0.5	0.5	0.5	0.5	0.5	0.25

5.3 SECURITY MONITORING

This section shows the results and computation of the benchmark score of the intruder detection application. The feature ratings for the intruder detection system are based on the criteria outlined in Table 2 in Section 4.3.1 and are discussed as follows:

- **Accuracy:** The accuracy of the intruder detection system is shown in Table 12.

Table 12: Intruder detection accuracy.

Area	Action	Intruder detected	Not detected
Living room	Walking	10	0
Living room	crouching	10	0
Living room	crawling	3	7
Bedroom	Walking	10	0
Bedroom	crouching	10	0
Bedroom	crawling	9	1
Kitchen	Walking	10	0
Kitchen	crouching	10	0
Kitchen	crawling	8	2

As the dining room and entrance were on the border of the sensor's coverage area, these are excluded from the calculations. From the table, the accuracy of the intruder detection system

was calculated to be 88.88%. Therefore, an (o) rating is given for this part.

- **Response time:** The response time of the intruder detection system is the same as the update rate of the radar platform, as no further processing is done with the data coming from the firmware. Hence, the rating given is (++).

The benchmark score for intruder detection is calculated based on the application ratings and feature scores for the radars from Table 3 and Table 4 in Section 4.3.2 and Section 4.3.3 respectively. Table 13 shows the combined numerical weights of the application and the feature score for the sensor and intruder detection system.

Table 13: Resolution (res), Update Rate (upd), Detection Range (det), Unobtrusiveness (uob), Processing Complexity (prc), Robustness (rob), Disturbance frequency (dfr), Calibration Complexity (cac), Unique limitation (unl), accuracy (acc), and response time (ret).

Features	res	upd	det	uob	prc	rob	dfr	cac	unl	acc	ret
Application Weight	1	0.75	1	1	0.25	0.75	0.5	0.75	0.25	1	0.5
Feature score	1	0.75	0.75	0.75	0.5	0.5	0.5	0.5	0.5	0.5	1

The equations 6 and 7 in Section 4.3.4 are used to compute the benchmark score (b_{s_l, a_j}) and normalized benchmark score ($b_{s_l, a_j, \text{norm}}$) for intruder detection application from the values in Table 13

$$b_{s_l, a_j} = 5.5625$$

$$b_{s_l, a_j, \text{norm}} = 0.6953$$

5.4 SOCIAL MONITORING

This section presents the results and computation of benchmark scores of the multiple people detection and visitor identification applications.

5.4.1 Multiple people detection

The feature ratings for the multiple people detection system are based on the criteria outlined in Table 2 in Section 4.3.1 and are discussed as follows:

- **Accuracy:** The accuracy of the multiple people detection system is shown in Table 14.

Table 14: Multiple people detection accuracy.

Area of the resident	Area of the visitor	Multiple people detected	Not detected
Living room	Living room	20	10
Living room	Bedroom	14	1
Living room	Kitchen	15	0
Bedroom	Living room	14	1
Bedroom	bedroom	17	13
Bedroom	Kitchen	15	0
Kitchen	Living room	15	0
Kitchen	Bedroom	15	0
Kitchen	Kitchen	30	0

As the dining room and entrance were on the borders of the sensor's coverage area, these are excluded from the calculations. From the table, the accuracy of the intruder detection system was calculated to be 86.11%. Therefore, an (o) rating is given for this part.

- **Response time:** Response time is the same as the update rate of the sensor platform, as no further processing is done. Hence, the rating given is (++).

The benchmark score for multiple people detection is calculated based on the application ratings and feature scores for the radars from Table 3 and Table 4 in Section 4.3.2 and Section 4.3.3 respectively. Table 15 shows the combined numerical weights of the application and the feature score for the sensor and multiple people detection system.

Table 15: Resolution (res), Update Rate (upd), Detection Range (det), Unobtrusiveness (uob), Processing Complexity (prc), Robustness (rob), Disturbance frequency (dfr), Calibration Complexity (cac), Unique limitation (unl), accuracy (acc), and response time (ret).

Features	res	upd	det	uob	prc	rob	dfr	cac	unl	acc	ret
Application Weight	0.5	0.5	0.75	0.75	0.5	0.75	0.5	0.75	0.25	0.75	0.5
Feature score	1	0.75	0.75	0.75	0.5	0.5	0.5	0.5	0.5	0.5	1

The equations 6 and 7 in Section 4.3.4 are used to compute the benchmark score (b_{s_l, a_j}) and normalized benchmark score ($b_{s_l, a_j, \text{norm}}$) for multiple people detection application from the values in Table 15

$$b_{s_l, a_j} = 4.25$$

$$b_{s_l, a_j, \text{norm}} = 0.6538$$

5.4.2 Visitor Identification

The assignment of the feature ratings for the visitor identification system was based on Table 2 and is discussed below:

- **Accuracy:** The confusion matrix of the visitor identification model on unseen test data is shown in Figure 30.

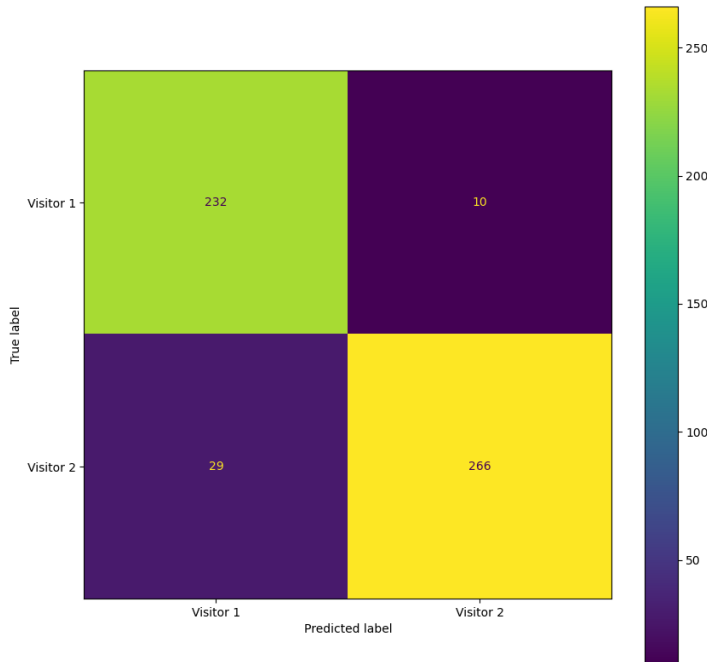


Figure 30: Confusion matrix for visitor identification model.

The system's performance was evaluated by calculating accuracy of the model using the Equation 9 mentioned in Section 4.4.1.2. The accuracy of this system was found to be 92.74%; hence, the rating for accuracy is given as (+).

- **Response time:** The average response time for predicting a visitor was 1.25 seconds. Therefore, the rating is given as (-).

The benchmark score for visitor identification is calculated based on the application ratings and feature scores for the radars from Table 3 and Table 4 in Section 4.3.2 and Section 4.3.3 respectively. Table 16 shows the combined numerical weights of the application and the feature score for the sensor and visitor identification system.

Table 16: Resolution (res), Update Rate (upd), Detection Range (det), Unobtrusiveness (uob), Processing Complexity (prc), Robustness (rob), Disturbance Frequency (dfr), Calibration Complexity (cac), Unique Limitation (unl), accuracy (acc), and response time (ret).

Features	res	upd	det	uob	prc	rob	dfr	cac	unl	acc	ret
Application Weight	1	0.75	0.75	0.75	0.25	0.75	0.5	0.75	0.25	0.75	0.5
Feature Score	1	0.75	0.75	0.75	0.5	0.5	0.5	0.5	0.5	0.75	0.25

The equations 6 and 7 in Section 4.3.4 are used to compute the benchmark score (b_{s_l, a_j}) and normalized benchmark score ($b_{s_l, a_j, \text{norm}}$) for visitor identification application from the values in Table 16

$$b_{s_l, a_j} = 4.625$$

$$b_{s_l, a_j, \text{norm}} = 0.6607$$

5.5 PHYSIOLOGICAL MONITORING

This section presents the results and computation of the benchmark score for the heart and breath rate detection applications. The assignment of feature ratings for the heart and breath rate system is based on Table 2 in Section 4.3.1 and is discussed below:

- **Accuracy:** To calculate the accuracy of both applications, ten readings from each sensor were taken, as shown in Tables 17 and 18.

Table 17: Heart rate accuracy.

Actual (bpm)	Measured (bpm)	Error (bpm)
68	57	11
68	61	7
72	70	2
71	67	4
71	64	7
69	55	14
72	50	22
69	60	9
75	64	11
70	56	14

Using the rating criteria mentioned in Table 6 from Section 4.4.5, a (-) rating was assigned as the average estimation error for the heart rate application was found to be 10.1 bpm.

Table 18: Breath rate accuracy.

Actual (bpm)	Measured (bpm)	Error (bpm)
31	13	18
26	15	11
27	17	10
27	11	16
27	22	5
28	21	7
26	13	13
27	15	12
26	19	7
26	13	13

Using the rating criteria mentioned in Table 6 from Section 4.4.5, a (-) rating was assigned as the average estimation error for the breath rate application was found to be 11.2 bpm.

Thus, the overall system has been assigned the same (-) rating, as both estimation errors fall within the 10 to 20 bpm range.

- **Response time:** The system required at least 20 seconds to stabilize readings for both heart and breath rates. Hence, a (-) rating is given for response time.

The benchmark score is calculated based on the application ratings and feature scores for the radars from Table 3 and Table 4 in Section 4.3.2 and Section 4.3.3 respectively. Table 19 shows the combined numerical weights of the application and the feature score for the sensor, plus the heart and breath rate system.

Table 19: Resolution (res), Update Rate (upd), Detection Range (det), Unobtrusiveness (uob), Processing Complexity (prc), Robustness (rob), Disturbance Frequency (dfr), Calibration Complexity (cac), Unique Limitation (unl), accuracy (acc), and response time (ret).

Features	res	upd	det	uob	prc	rob	dfr	cac	unl	acc	ret
Application Weight	1	0.75	1	0.75	0.5	0.75	0.75	0.75	0.75	0.25	0.5
Feature Score	1	0.75	0.75	0.75	0.5	0.5	0.5	0.5	0.5	0.25	0

The equations 6 and 7 in Section 4.3.4 are used to compute the benchmark score (b_{s_l, a_j}) and normalized benchmark score ($b_{s_l, a_j, \text{norm}}$) for the heart and breath rate application from the values in Table 19.

$$b_{s_l, a_j} = 4.6875$$

$$b_{s_l, a_j, \text{norm}} = 0.6048$$

DISCUSSION

This chapter discusses the strengths and limitations observed while evaluating the radar systems implemented for various smart home health monitoring categories. A comparison of the benchmark scores of the applications is also discussed.

The research started with the study of three radar platforms from Seeed Studios (Human static presence module Lite (MR24HPC₁), Fall detection module (MR24FDB₁), and Respiratory sleep detection module (MR24BSD₁)), which were available in the HINT environment. It was understood that these sensor platforms were designed for specific applications like human static presence detection, fall detection, and respiratory and sleep detection. The output data coming from these platforms was limited to these applications as all processing was done on the board. Radar point cloud data, which gives information about the position, velocity, and orientation of detected targets, couldn't be retrieved from these radar platforms. Hence, these sensors were not suitable for the applications to be implemented in this thesis.

Recent works that use radars for different applications have been shown to use radars from suppliers like TI and Infineon. In the comparative study of sensors for indoor monitoring done by Piroso [30], which used TI IWR6843ISK and Infineon Position2go radar platforms, it was found that the TI IWR6843ISK radar sensor is capable of tracking and localization of humans in indoor environments. The Infineon Position2go sensor requires additional processing, such as obtaining a moving average of the incoming values, which can affect the response time and performance of the system. However, raw radar data can be recorded from the position2go sensor, whereas in IWR6843ISK, additional hardware equipment was required to extract raw data. Signal processing techniques are done on this raw data to implement application-specific functionalities. Since this thesis is on the feasibility study of radar systems for various health monitoring services and the firmware of IWR6843ISK gave rich information for implementing the health monitoring systems, extracting raw data was not required.

The data processing of the output data from the IWR6843ISK firmware was done using an open-source Python script. This approach simplified data handling and analysis, which became the backbone of the evaluation experiments in the thesis. All the experiments for the evaluations were done using just one unit of the IWR6843ISK. The results and discussions are based on this unit. The results might or might not be similar when a different unit of the same sensor is used.

6.1 LOCALIZATION

The localization application utilizes the 3D people counting demo firmware data directly. The discussion of the results is based on the firmware's output. The mean absolute error of the radar's localization system was 17.39 cm when the target was within its field of view inside the home environment. The localization errors for the related works were 69.74 cm in [31] and 42 cm in [32]. The related works were different regarding the environment, positioning, and experiments conducted, so a direct comparison is not justified. However, it can be stated that the localization error achieved by the system is considerably low in an indoor environment.

The sensor's real-time update rate ensured continuous and effective tracking of resident's movement. Even though the radar's detection range is good, it couldn't cover the entire HINT, as the signals cannot penetrate through walls. The regions behind doors and walls, such as the bathroom and areas in the bedroom behind the walls, couldn't be covered. This can be seen in the coverage plot in Figure 28. This shows that mmWave readers are feasible for the localization application. The limitations of the coverage area can be addressed by integrating multiple radars.

The accuracy of the firmware decreased drastically in the dining room and entrance areas, which are in the extreme borders of the radar's Field of View (FoV). Even though the azimuth angle of the radar's FoV is 120 degrees, the localization accuracy is low towards the borders. Since only approximate accuracy is required for smart home applications, the TI IWR6843ISK radar module with the 3D people counting demo firmware is feasible for non-obtrusive tracking and localization of elderly people living alone. However, more than one radar sensor is required for larger homes or homes with many rooms.

6.2 HUMAN ACTIVITY RECOGNITION

The application works on the point cloud data from the 3D people counting demo firmware. The discussion of the results is based on the deep learning model implemented in this thesis. The overall accuracy of the implemented HAR system was 97.80 %, while the accuracy for the independent activities of walking, standing, and sitting were 96.84%, 97.81%, and 100%, respectively. Comparison of these results directly with the related works is not justifiable due to the difference in type of activities, number of test subjects, radar data processing, and environment. The accuracy for standing and sitting are 62.2%, and 52.7% in [33]; the results are low as this approach didn't use any DL models for the classification. The highest accuracy received for the CNN + LSTM model in [35] for the activities walking away, walking

towards, sitting down, and standing up are 94%, 87%, 63%, and 58%, respectively. These results were achieved on the dataset collected in a new environment with increased clutter. The other work that used an end-to-end convolution model received an accuracy of 100% and 94% for sitting still and walking, respectively [13]. This method classifies a total of 7 actions, with an average accuracy of 96.3%. Although the scores of this model are much better than the previous methods, the computational requirements are much higher because of the increase in deep learning parameters.

The implemented system also predicted the activities in real-time. This shows that mmWave radar can identify and classify basic human activities in a home environment. The radar's positioning influenced the system's accuracy, as activities occurring close or at the edges of the radar's field of view were often misclassified or missed entirely. The system needs a lot of training data with corresponding labels for the deep learning model. Different people perform actions differently. The indoor environment in which the model is trained will bias the output of the model. The HAR model using mmWave radars trained in one environment performs poorer in a different unseen environment, as shown in [35]. The generalization capabilities of HAR models using mmWave radars are low.

The point cloud data of the firmware cannot distinguish between different individuals present in the FoV. Hence, the implemented system was trained on a dataset collected when a single person was in the vicinity of the radar, so it only classifies the actions of one person. If multiple visitors are present, the system classifies the action of the person closest to the radar.

In a real home environment, people tend to do more complex activities than sitting, standing, and walking. These aspects need to be included in a system to monitor people properly. More training data and a better model are required for recognizing these activities. Additionally, in real-time, humans perform actions concurrently, whereas the data is recorded in a few samples of time. This is another challenge when implementing more complex HAR systems.

The evaluation shows that mmWave radars are feasible for the HAR application. However, it has generalization and coverage issues. Using multiple radars can address the coverage issues. The 3D people counting demo firmware is not feasible to classify the activities of multiple people in the smart home. The implemented HAR system has the limitation of only classifying 3 basic activities. Hence, it is not feasible in a real smart home where older adults are living.

6.3 FALL DETECTION

Fall detection is a critical application in health monitoring for elderly people. The height data coming from the 3D people counting demo

firmware is used to implement the fall detection application. The discussion of the results is based on the fall detection system implemented in this thesis. The system successfully identified most simulated falls and exhibited low false positive rates for non-fall events like sitting or bending down, as outlined in Table 10. The total accuracy of the implemented method is 90.00%, which is higher than the scores achieved in related works [40] and [33]. This direct comparison is also not justified as the methods used different numbers of test subjects and trials.

However, in [41], using a deep learning model that combines two radars achieved an accuracy in the range of 90-98%, depending on the positioning. This proved to be a much better approach than the implemented method. The use of deep learning models can increase the accuracy of the fall detection system, but it can also increase the computational requirements. Most publicly available fall datasets consist of data from young or middle-aged adults. The pattern of falls for such adults might be different compared to an elderly person, which may create challenges in accurately detecting falls in the elderly.

The placement of the radar played a crucial role in its performance, as the location and layout of the environment influenced its effectiveness. Special cases were considered in the kitchen area to detect falls and also to identify falls happening behind furniture in other areas. Including such exceptional cases is not practical in terms of adaptability and generalization. The implemented system also needed tuning of parameters, such as the fall height percentage and the number of frames. These parameters may change depending on the home environment, positioning of the radar, and type of individuals. The system was not able to detect the falls in the dining and entrance areas despite the special conditions. This is because of the limitation of the mmWave radar's field of view.

The response time of the implemented system is 3 seconds, which was added to reduce false positive cases.

The simulated falls for evaluating the systems were based on assumptions about what a fall might look like, but actual falls can vary significantly between individuals. The variability in how different individuals fall—whether quickly or slowly, depending on the person and their physical condition could lead to improper detection of falls. Elderly people's falls might differ entirely from the simulated falls conducted in the experiments due to the difference in age and physical conditions. Hence, even though the radar-based system was able to detect most of the falls, it cannot be deployed in an actual home environment to monitor the safety of elderly people living alone. The evaluation shows that it is feasible to use mmWave radars for fall detection applications, provided the entire home is covered by integrating multiple radar units. However, the implemented system based on

the firmware's data is not feasible for a smart home for the safety of elderly people.

6.4 INTRUDER DETECTION

The intruder detection application utilizes the 3D people counting demo firmware data directly. The discussion of the results is based on the firmware's output. The radar system demonstrated an accuracy rate of 88.88% in detecting intruders entering the home through the designated entry points when the resident was away. However, it is essential to note that this detection rate cannot be directly compared to other systems, as the methodologies and experiments used for the evaluation differed from those used in other systems. However, this system has limitations in detecting intruders in areas outside of its radar's FoV, such as certain parts of the dining room and entrance. Given the radar's narrow vertical viewing angle of ± 15 degrees, it sometimes failed to detect an intruder approaching in a crawling position. So, if an intruder breaks in through either of the two doors and remains in the radar's blind spot, the system won't be able to detect the intruder.

The system also didn't detect the movements of the vacuum cleaner robot as an intruder. This shows that the firmware is optimized for detecting the movements of people or large objects. The robot's movement is similar to that of small pets, so the system might not falsely detect small pets as intruders when the resident is not home. However, slightly larger pets might be detected as intruders, which can be a limitation.

The system is designed in such a way that the resident has to manually switch on/off the security mode. This might be inconvenient for elderly people, as they might forget to switch to security mode when they are leaving the home or switch off when they are entering. Both these actions might lead to the system not detecting or falsely detecting intruders. This is the limitation of the implemented system and is not caused by radar's features.

Overall, the radar system demonstrated strong capabilities in detecting intruders from both entry points and shows that mmWave radars are feasible for intruder detection applications. However, multiple radar units need to be integrated to get full coverage of the smart home. The radar-based system is feasible in a smart home for elderly people if automatic switching of security mode is implemented.

6.5 MULTIPLE PEOPLE DETECTION

Multiple people detection applications help monitor the in-person social activity of older adults living alone in their homes. The multiple people detection in this thesis utilizes the 3D people counting

demo firmware data directly. The discussion of the results is based on the firmware's output. The radar system demonstrated an accuracy of 86.11% to detect multiple people within the monitored space. This system's performance also cannot be directly compared to the related works as the methods implemented are entirely different. The response time is similar to the localization system, as the data from the firmware is directly used. One significant issue was the occurrence of ghost targets or false detection. Experiments were conducted when the resident was alone in the HINT, and a refrigerator door or a shelf was opened and closed. The firmware detected this as a person, causing false detection of multiple people. When the resident walked alone, ghost targets were detected in the bedroom area outside the radar's line of sight. However, this occurrence of ghost targets was very rare. These are the limitations of the firmware.

Occlusions also caused challenges in detecting multiple people. When the visitor and the resident are standing in the same line as that of the radar's field of view, the system identifies them as a single person. The firmware doesn't distinguish between two people if they are of similar heights and standing close to each other. This issue was particularly evident in more confined areas, as shown in Table 14. When the resident and visitor were in the same room, the likelihood of occlusion was high, leading to reduced accuracy in detecting multiple individuals. However, the effect of occlusion was less in the kitchen, where the entire area was fully covered under the radar's line of sight. Even though one person was standing behind another, the firmware was able to detect each of them separately. The effect of occlusion is common to all mmWave radars and is a limitation to most of the applications.

Overall, the system effectively detected multiple people, showing that mmWave radars are feasible for the application to monitor the social activities of older adults living alone. Integration of multiple radars can solve the coverage issues and the effects of occlusions. The firmware is not feasible for multiple people detection in a smart home for healthcare applications due to the appearance of ghost targets and poor detection results.

6.6 VISITOR IDENTIFICATION

The application works on the point cloud data from the 3D people counting demo firmware to detect the gait patterns of visitors. The discussion of the results is based on the deep learning model implemented in this thesis. The radar system was able to identify two different visitors based on their gait patterns, and the system was able to predict in real-time with a delay of only a few seconds. The implemented system achieved an accuracy of 92.74% in identifying the two visitors. The implemented system cannot be compared directly

with the related works, as the gait recognition models were classifying more people after utilizing more training data. The method in [47] achieved an accuracy of 88.59% in identifying five different subjects, whereas the method in [48] reported an accuracy of 96.7% for classifying 40 subjects. The system can properly classify the visitors in the center of the FoV of the sensor.

The deep learning models required extensive data collection to recognize each person's gait, which proved time-consuming and resource-intensive. Experiments were conducted with two visitors who had different physiques. Despite this, a significant amount of data was needed to distinguish between them. The firmware's output of point cloud data doesn't differentiate multiple people in a frame, so data was recorded only when a single person was in the field of view of the radar. Thus, the system can only identify a single person in the radar's field of view. In a practical scenario, multiple people might be present. The method proposed in [48] was able to classify with an accuracy of 90.2% in multiple people scenarios, showing the potential of using radars for this application. However, it might be difficult and inconvenient to record and label the data of regular visitors to the homes of elderly people.

The evaluation shows that mmWave radar is feasible for visitor identification applications in a smart home. The integration of multiple radars can help increase the coverage inside the smart home. The implemented model based on the firmware data is not feasible as it can only classify one person in the vicinity.

6.7 HEART AND BREATH RATE DETECTION

The 3d people counting demo firmware is used to detect vital signs such as a human's heart and breath rate. The results and discussion are based on the output of the firmware. The average estimation error for the heart rate and breath rate application was 10.1 bpm and 11.2 bpm respectively. The readings of the radar were always lower compared to the readings of the reference sensor, as shown in Table 17 and 18.

To get stable readings from the firmware, the individual had to remain stationary for at least 20 seconds. Any movement during the detection period caused fluctuations in readings, which compromised the accuracy of the vital sign measurements. The breathing of the person being monitored had to be deep enough to induce chest movements alone. If there were minimal or no chest movements, the radar didn't detect the target at all. Even though the normal breath rate of adults is 12-20 per minute [63, 64]. These deep breaths for the experiments resulted in high values of the breath rate shown in the reference sensor. The person had to sit still and maintain chest movements for the firmware to give stable readings for the heart rate and breath

rate. The values considered for computing the estimation error were also taken when the radar gave stable readings. Hence, this system cannot be used for direct comparison with other systems that detect heart and breath rates.

The main goal of health monitoring systems is to provide continuous monitoring of older adults living alone and provide assistance when required. It is impractical to expect or request an elderly person living alone to sit still for 20 seconds and breathe in front of the radar frequently to get updates on their vital signs. Even though the firmware can detect vital signs up to a range of 5 meters, the accuracy keeps reducing with the increase in distance. Additionally, the firmware cannot detect the vital signs of multiple individuals, which further limits its applicability in settings where more than one person is present in the radar's vicinity. All these limitations of the firmware to get a stable measurement for heart rate and breath rate make it infeasible to deploy in a smart home to monitor the vital signs of an elderly person. Although the evaluated system based on the firmware is not feasible, mmWave radar technology is feasible for physiological monitoring, as can be seen in the literature survey.

6.8 BENCHMARK SCORES

The benchmark score provides quantitative insights into the radar system's performance across these seven different smart home monitoring applications. The Benchmark scores attained for each of the applications are mentioned in Table 20.

Table 20: Benchmark scores of the distinct applications.

Application	Benchmark score
Localization	0.7142
Human activity recognition	0.7250
Fall detection	0.6532
Intruder detection	0.6953
Multiple people detection	0.6538
Visitor identification	0.6607
Heart and breath rate detection	0.6048

The benchmark score, as such, doesn't provide much information. Ideally, the scores should range from 0 to 1. According to the central bias tendency, as mentioned in section 4.3.4, the scores tend to cluster around an average of 0.5. To make the scores more readable, correc-

tion is applied using Equation 8, with $\alpha = 10$ for getting significant differences in scores. The modified benchmark scores are calculated as shown in Table 21

Table 21: Modified benchmark scores of the distinct applications.

Application	Modified benchmark score ($\alpha = 10$)
Localization	6.97
Human activity recognition	7.61
Fall detection	4.16
Intruder detection	5.95
Multiple people detection	4.18
Visitor identification	4.44
Heart and breath rate detection	2.71

The weight assignment for the features of a particular application is subjective to authors [3]. This is a major limitation of the model since comparison can be inaccurate if different authors assign weights differently to a single application. To address this limitation, a survey was conducted with a brief discussion about the evaluation methods, followed by questionnaires of the participants' agreement with the application weights provided in this thesis. Options are provided for the participants to offer different weights and rating criteria. The survey results showed that the majority of the participants agreed with the weights assigned and with the criteria employed for the benchmark model.

Provided that the same feature weights are assigned, this benchmark model can be used to compare sensor-based systems designed for specific applications. For instance, it can compare the same type of sensor module from different vendors that are implemented for the same application. Similarly, it can be used to compare entirely different sensor types and their systems designed for a specific application. Furthermore, the model is also suitable for comparing sensor-based systems that perform different applications, ensuring that diverse technologies and their capabilities can be accurately evaluated for their intended purposes.

From Table 21, The scores of the human activity recognition, localization, and intruder detection applications are in the top 3 positions, indicating that these applications are most suitable for the radar-based system in this smart home environment.

Although human activity recognition has the highest benchmark score, it is not recommended for practical deployment. This is because the model only considers three activities and relies on a deep learning model, which may not generalize well to more complex real-world environments. Localization with this radar-based system can be deployed in a practical environment if there is full coverage in the environment. Intruder detection with this radar-based system, while ranked third, poses challenges for actual household implementation, mainly due to the need for manual activation and deactivation, which is a significant issue in homes with elderly residents who may struggle to remember to switch modes consistently.

Visitor identification, multiple people detection, and fall detection receive the next best scores and these are in similar ranges. This shows that these radar-based systems need improvement in accuracy or response time, which led to a decrease in the overall benchmark scores.

Heart and breath rate detection is the application with the lowest score, indicating that this is the least suitable application compared with the other evaluated systems. This implementation requires significant improvements to be viable for real-world deployment.

CONCLUSION

This chapter concludes with the inference from the experiments conducted to evaluate the mmWave radar-based smart home monitoring systems for the healthcare of older adults. The chapter also discusses potential future work.

The thesis used TI's IWR6843ISK radar platform to evaluate radar-based health monitoring systems for older adults living alone. Seven different applications were implemented for a smart home's distinct categories of health monitoring systems. The data from the firmware provided by TI was directly used to implement these applications: "3D people counting demo" and "vital signs with people tracking." "The 3D people counting demo" was used for the following applications: Localization, HAR, fall detection, intruder detection, multiple people detection, and visitor identification. "The vital signs and people tracking" was used to detect the heart and breath rate.

The evaluation of the mmWave radar system showed that it is feasible in smart home monitoring systems for older adults as a non-invasive, contactless solution. This technology offers promising potential for improving healthcare outcomes and enhancing the independence of elderly individuals. Through the experiments, it was found that all the applications suffered common limitations due to the lack of coverage area of a single radar unit. The performance of the radar deteriorated towards the borders of its field of view. The inability of the mmWave radar to penetrate thick walls also limited the coverage area in the environment. Another limitation of mmWave radars was occlusion, which disrupted continuous monitoring. Even though the experiments were done using TI IWR6843ISK, these results are common to all mmWave radars.

The evaluation of the systems showed that firmware "3D people counting demo" is feasible for localizing and intruder detection applications in a smart home as it provided good benchmark scores in the radar's field of view. The firmware is not feasible for multiple people detection due to false positive detections by the firmware. The firmware's inability to differentiate the point clouds of different individuals makes it infeasible for visitor identification and HAR. The firmware "The vital signs and people tracking" is not feasible for heart and breath rate detection in smart home health care applications as the readings were unstable.

The implemented system for fall detection is not feasible as it provides low accuracy and response time. The HAR system considered only three basic activities; hence, it is not feasible in practical sce-

narios. The HAR and Visitor identification systems were based on the deep learning models that used the point cloud data from the firmware. There are generalization issues while using such models, such as the difference in the positioning of the radar sensor and the environment influencing the model's performance. Datasets used commonly for such applications lack standardization due to variations in how individuals perform activities, influenced by age, height, and weight.

The benchmark model used to compare each radar-based application has advantages and limitations. The updated model can be used to compare the sensor-based system by including its features. This model enables practical evaluation of the entire system along with the features of the sensor. The benchmark scores can be used to compare other implementations of radar-based or other sensor-based systems for corresponding applications. However, the models' weight assignment to the features of an application being subjective to different authors is a drawback. This thesis used a survey to back the rating criteria and the assigned application weights. Most of the participants who participated in the survey agreed with the proposed weights and ratings.

In future work, the evaluation of integrating multiple radars for more coverage and continuous and seamless health monitoring services. Developing custom firmware that uses the radar's raw data to process and tune separately for each application to yield better benchmark scores. For example, the point cloud of multiple targets can be extracted separately to distinguish visitors and residents and recognize the activities of each individual, thus enhancing visitor identification and HAR.

Different implementations of the applications can be compared with the basic implementation done in this thesis using the modified benchmark model, provided the authors agree with the feature weights assigned for the applications. The updated model can also be used to compare other sensor-based systems. The mmWave radar signal's ability to penetrate through materials with less thickness and density can be utilized to hide the entire system behind these materials, making them even more non-obtrusive and non-invasive.

While mmWave radar technology is feasible for smart home monitoring systems in healthcare applications, there are limitations in the implemented systems that require further research and development.

BIBLIOGRAPHY

- [1] C. Iovescu and S. Rao, "The fundamentals of millimeter wave sensors," *Texas Instruments*, pp. 1–8, 2017.
- [2] Texas Instruments, "Texas instruments radar toolbox." [Online]. Available: https://dev.ti.com/tirex/explore/node?a=1AslXXD_1.30.01.03&node=A__AI4TjoY4aHFIgDAH2Mcpyw__radar_toolbox__1AslXXD_LATEST&placeholder=true
- [3] A. Braun, R. Wichert, A. Kuijper, and D. W. Fellner, "A benchmarking model for sensors in smart environments," in *Ambient Intelligence: European Conference, Aml 2014, Eindhoven, The Netherlands, November 11-13, 2014. Revised Selected Papers 5*. Springer, 2014, pp. 242–257.
- [4] D. J. Cook and S. K. Das, "How smart are our environments? an updated look at the state of the art," *Pervasive and Mobile Computing*, vol. 3, pp. 53–73, 03 2007.
- [5] G. Bhavanasi, L. Werthen-Brabants, T. Dhaene, and I. Couckuyt, "Patient activity recognition using radar sensors and machine learning," *Neural Computing and Applications*, vol. 34, pp. 16 033–16 048, 05 2022.
- [6] A. Brush, M. Hazas, and J. Albrecht, "Smart homes: Undeniable reality or always just around the corner?" *IEEE Pervasive Computing*, vol. 17, pp. 82–86, 01 2018.
- [7] B. K. Hensel and G. Demiris, "Technologies for an aging society: A systematic review of "smart home" applications," *Yearbook of Medical Informatics*, vol. 17, pp. 33–40, 08 2008.
- [8] S. A. Shah and F. Fioranelli, "Rf sensing technologies for assisted daily living in healthcare: A comprehensive review," *IEEE Aerospace and Electronic Systems Magazine*, vol. 34, pp. 26–44, 11 2019.
- [9] G. Cicirelli, R. Marani, A. Petitti, A. Milella, and T. D'Orazio, "Ambient assisted living: a review of technologies, methodologies and future perspectives for healthy aging of population," *Sensors*, vol. 21, no. 10, p. 3549, 2021.
- [10] M. Espinilla, L. Martinez, J. Medina, and C. Nugent, "The experience of developing the ujami smart lab," *IEEE Access*, vol. 6, pp. 34 631–34 642, 2018.

- [11] F. S. Lesani, F. Fotouhi Ghazvini, and H. Amirkhani, "Smart home resident identification based on behavioral patterns using ambient sensors," *Personal and Ubiquitous Computing*, vol. 25, pp. 151–162, 2021.
- [12] Z. Shen, J. Nunez-Yanez, and N. Dahnoun, "Multiple human tracking and fall detection real-time system using millimeter-wave radar and data fusion," *2023 12th Mediterranean Conference on Embedded Computing (MECO)*, 06 2023.
- [13] W. Ye, H. Chen, and B. Li, "Using an end-to-end convolutional network on radar signal for human activity classification," *IEEE Sensors Journal*, vol. 19, pp. 12 244–12 252, 12 2019.
- [14] T. Instrument, "How mmwave sensors create technology advantages for independent, "assisted" living." [Online]. Available: <https://www.ti.com/lit/ta/sszt103/sszt103.pdf?ts=1717402232347>
- [15] J. A. Álvarez-García, P. Barsocchi, S. Chessa, and D. Salvi, "Evaluation of localization and activity recognition systems for ambient assisted living: The experience of the 2012 eval competition," *Journal of Ambient Intelligence and Smart Environments*, vol. 5, no. 1, pp. 119–132, 2013.
- [16] M. I. Skolnik, *Introduction to radar systems*. Mcgraw-Hill, 2001.
- [17] M. A. Richards, J. Scheer, W. A. Holm, and W. L. Melvin, "Principles of modern radar," 2010.
- [18] S. Rao, "Introduction to mmwave sensing: Fmcw radars," *Texas Instruments (TI) mmWave Training Series*, pp. 1–11, 2017.
- [19] R. Ferguson, M. Chevrier, and A. Rankin, "mmwave radar: Enabling greater intelligent autonomy at the edge," *Texas Instruments*, pp. 1–9, 2018.
- [20] J. Zhang, R. Xi, Y. He, Y. Sun, X. Guo, W. Wang, X. Na, Y. Liu, Z. Shi, and T. Gu, "A survey of mmwave-based human sensing: Technology, platforms and applications," *IEEE Communications Surveys & Tutorials*, 2023.
- [21] M. Steinhauer, H.-O. Ruoss, H. Irion, and W. Menzel, "Millimeter-wave-radar sensor based on a transceiver array for automotive applications," *IEEE Transactions on Microwave Theory and Techniques*, vol. 56, pp. 261–269, 02 2008.
- [22] C. Vasanelli, R. Batra, A. D. Serio, F. Boegelsack, and C. Waldschmidt, "Assessment of a millimeter-wave antenna system for mimo radar applications," *IEEE Antennas and Wireless Propagation Letters*, vol. 16, pp. 1261–1264, 2017.

- [23] Digi-Key Electronics. (Year) Radar-based systems aid the change from smart buildings to homes that care. [Online]. Available: <https://www.digikey.se/en/blog/radar-based-systems-aid-the-change-from-smart-buildings-to-homes-that-care>
- [24] M. Stephan and A. Santra, "Radar-based human target detection using deep residual u-net for smart home applications," *International Conference on Machine Learning and Applications*, 12 2019.
- [25] A. Santra, R. V. Ulaganathan, and T. Finke, "Short-range millimetric-wave radar system for occupancy sensing application," *IEEE Sensors Letters*, vol. 2, pp. 1–4, 09 2018.
- [26] J. Wu, H. Cui, and N. Dahnoun, "A novel high performance human detection, tracking and alarm system based on millimeter-wave radar," 06 2021.
- [27] M. Jiang, S. Guo, H. Luo, and G. Cui, "Continuous tracking of indoor human targets based on millimeter wave radar," *2022 Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC)*, 11 2022.
- [28] H. Cui and N. Dahnoun, "High precision human detection and tracking using millimeter-wave radars," *IEEE Aerospace and Electronic Systems Magazine*, vol. 36, pp. 22–32, 01 2021. [Online]. Available: https://research-information.bris.ac.uk/ws/portalfiles/portal/248162747/two_radar_final.pdf
- [29] W. Li, Y. Wu, R. Chen, H. Zhou, and Y. Yu, "Indoor multihuman device-free tracking system using multiradar cooperative sensing," *IEEE Sensors Journal*, vol. 23, pp. 27 862–27 871, 11 2023.
- [30] C. Piroso, "Comparison of radar sensors for indoor human localization," Master's thesis, Politecnico di Torino, 2022.
- [31] X. Wang, Z. Zhang, N. Zhao, Y. Zhang, and D. Huang, "Indoor localization and trajectory tracking system based on millimeter-wave radar sensor," in *2021 IEEE 10th Data Driven Control and Learning Systems Conference (DDCLS)*, 2021, pp. 1141–1147.
- [32] W. Li, R. Chen, Y. Wu, and H. Zhou, "Indoor positioning system using a single-chip millimeter wave radar," *IEEE Sensors Journal*, vol. 23, no. 5, pp. 5232–5242, 2023.
- [33] T. Johansson and L. Wikström, "An exploratory study of simple fall and activity recognition using mmwave," 2021.
- [34] M. M. Islam, S. Nooruddin, F. Karray, and G. Muhammad, "Human activity recognition using tools of convolutional neural networks: A state of the art review, data sets, challenges, and future prospects," *Computers in Biology and Medicine*, vol. 149, p. 106060, 10 2022.

- [35] H.-U.-R. Khalid, A. Gorji, A. Bourdoux, S. Pollin, and H. Sahli, "Multi-view cnn-lstm architecture for radar-based human activity recognition," *IEEE Access*, vol. 10, pp. 24 509–24 519, 2022.
- [36] Y. Kim and T. Moon, "Human detection and activity classification based on micro-doppler signatures using deep convolutional neural networks," *IEEE geoscience and remote sensing letters*, vol. 13, no. 1, pp. 8–12, 2015.
- [37] L. Z. Rubenstein, "Falls in older people: epidemiology, risk factors and strategies for prevention," *Age and ageing*, vol. 35, no. suppl_2, pp. ii37–ii41, 2006.
- [38] A. Bergland and K. Laake, "Concurrent and predictive validity of "getting up from lying on the floor"," *Aging clinical and experimental research*, vol. 17, pp. 181–185, 2005.
- [39] F. Bloch, "Critical falls: why remaining on the ground after a fall can be dangerous, whatever the fall," *Journal of the American Geriatrics Society*, vol. 60, no. 7, pp. 1375–1376, 2012.
- [40] K. Wang, G. Zhan, and W. Chen, "A new approach for iot-based fall detection system using commodity mmwave sensors," in *Proceedings of the 2019 7th International Conference on Information Technology: IoT and Smart City*, 2019, pp. 197–201.
- [41] K. Raskov and O. Christiansson, "Machine learning for radar in health applications: Using machine learning with multiple radars to enhance fall detection," 2022.
- [42] A. Okar and C. Wu, "Feasibility study of millimeter wave radars for safeguards applications," Brookhaven National Laboratory (BNL), Tech. Rep. BNL-223089-2022-INRE, 2022, u.S. Department of Energy Office of Scientific and Technical Information. [Online]. Available: <https://www.osti.gov/servlets/purl/1896881>
- [43] V. G. S., B. Pardhasaradhi, and P. Srihari, "Intruder detection and tracking using 77ghz fmcw radar and camera data," in *2020 International Radar Symposium (IRS)*. IEEE, 2020, pp. 1–6.
- [44] S. Ashida and C. A. Heaney, "Differential associations of social support and social connectedness with structural features of social networks and the health status of older adults," *Journal of aging and health*, vol. 20, no. 7, pp. 872–893, 2008.
- [45] M. E. Morris, B. Adair, E. Ozanne, W. Kurowski, K. J. Miller, A. J. Pearce, N. Santamaria, M. Long, C. Ventura, and C. M. Said, "Smart technologies to enhance social connectedness in older people who live at home," *Australasian journal on ageing*, vol. 33, no. 3, pp. 142–152, 2014.

- [46] A. Gaete, F. J. Gutierrez, S. F. Ochoa, P. Guerrero, and A. Wyzykowski, "Visitrack: A pervasive service for monitoring the social activity of older adults living at home," in *Ubiquitous Computing and Ambient Intelligence: 11th International Conference, UCAmI 2017, Philadelphia, PA, USA, November 7–10, 2017, Proceedings*. Springer, 2017, pp. 520–530.
- [47] Z. Ni and B. Huang, "Gait-based person identification and intruder detection using mm-wave sensing in multi-person scenario," *IEEE Sensors Journal*, vol. 22, no. 10, pp. 9713–9723, 2022.
- [48] T. Li, Z. Zhao, Y. Luo, B. Ruan, D. Peng, L. Cheng, C. Shi *et al.*, "Gait recognition using spatio-temporal information of 3d point cloud via millimeter wave radar," *Wireless Communications and Mobile Computing*, vol. 2022, 2022.
- [49] Z. Yang, P. H. Pathak, Y. Zeng, X. Liran, and P. Mohapatra, "Monitoring vital signs using millimeter wave," in *Proceedings of the 17th ACM international symposium on mobile ad hoc networking and computing*, 2016, pp. 211–220.
- [50] K. Guo, C. Liu, S. Zhao, J. Lu, S. Zhang, and H. Yang, "Design of a millimeter-wave radar remote monitoring system for the elderly living alone using wifi communication," *Sensors*, vol. 21, p. 7893, 11 2021.
- [51] A. Ahmad, J. C. Roh, D. Wang, and A. Dubey, "Vital signs monitoring of multiple people using a fmcw millimeter-wave sensor," in *2018 IEEE Radar Conference (RadarConf18)*. IEEE, 2018, pp. 1450–1455.
- [52] P. Zhao, C. X. Lu, B. Wang, C. Chen, L. Xie, M. Wang, N. Trigoni, and A. Markham, "Heart rate sensing with a robot mounted mmwave radar," in *2020 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2020, pp. 2812–2818.
- [53] J. Lundström, W. O. De Moraes, M. Menezes, C. Gabrielli, J. Bentes, A. Sant'Anna, J. Synnott, and C. Nugent, "Halmstad intelligent home-capabilities and opportunities," in *Internet of Things Technologies for HealthCare: Third International Conference, HealthyIoT 2016, Västerås, Sweden, October 18–19, 2016, Revised Selected Papers 3*. Springer, 2016, pp. 9–15.
- [54] T. Instrument, "Firmwares supported for each radar device - overview." [Online]. Available: https://dev.ti.com/tirex/explore/node?node=A__ALaoNkZs5b5KkCoeLjooEg__radar_toolbox__1AslXXD__LATEST&placeholder=true
- [55] T. Instruments, "3d people counting demo software implementation guide -rev 2.3," 2021. [Online]. Available: https://dev.ti.com/tirex/explore/node?node=A_

- [_AOoElvaXdtFqyo2XLLheCQ__radar_toolbox__1AslXXD__LATEST&placeholder=true](#)
- [56] —, “Detection layer parameter tuning guide for the 3d people counting demo revision 1.0,” 2021. [Online]. Available: https://e2e.ti.com/cfs-file/__key/communityserver-discussions-components-files/1023/3D_5Foo_people_5Foo_counting_5Foo_detection_5Foo_layer_5Foo_tuning_5Foo_guide.pdf
- [57] Texas Instruments, “Group tracker parameter tuning guide for the 3d people counting demo -rev 1.0.” [Online]. Available: https://e2e.ti.com/cfs-file/__key/communityserver-discussions-components-files/1023/3D_5Foo_people_5Foo_counting_5Foo_tracker_5Foo_layer_5Foo_tuning_5Foo_guide.pdf
- [58] T. Instrument, “Ti radar toolbox 4.11.0 : Vital signs with people tracking,” 2024, accessed: 2024-07-29. [Online]. Available: https://dev.ti.com/tirex/explore/node?a=VLyFKFf__4.12.0&node=A__ANQFjINyZ2YdSLVBPxylNw__com.ti.mmwave_industrial_toolbox__VLyFKFf__4.12.0&r=VLyFKFf__4.11.0
- [59] A. Braun, “Application and validation of capacitive proximity sensing systems in smart environments,” *Journal of Ambient Intelligence and Smart Environments*, vol. 7, no. 5, pp. 693–694, 2015, refer to chapter Benchmarking model for sensors in smart environments.
- [60] L. E. Crawford, J. Huttenlocher, and P. H. Engbretson, “Category effects on estimates of stimuli: Perception or reconstruction?” *Psychological Science*, vol. 11, no. 4, pp. 280–284, 2000.
- [61] M. Hitka, M. Gejdoš, and M. Sydor, “Dataset on anthropometric measurements of the adult population in slovakia,” *Data in Brief*, vol. 55, p. 110536, 2024. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2352340924005031>
- [62] Nonin, “Onyx 3 product specification sheet,” 2018. [Online]. Available: <https://www.nonin.com/wp-content/uploads/2018/09/Onyx-III-9591-Product-Spec-Sheet.pdf>
- [63] K. L. McCance and S. E. Huether, *Pathophysiology: The biologic basis for disease in adults and children*. Mosby/Elsevier, 2014. [Online]. Available: <https://books.google.com>
- [64] M. Walsh, G. Erb, and B. Kozier, *Fundamentals of Nursing: Frenchs Forest*. NSW, Australia: Pearson Education, 2010.

APPENDIX

This section of the appendix chapter provides the detailed tables that support the consolidated summaries presented in the results Chapter 5 of this thesis.

Table 22: Mean absolute error for points measured in the bedroom area of HINT

Area	Point	Actual X	Actual Y	Measured X	Measured Y	Error (in cm)
Bedroom	Point 1	-23.00	346.00	-17.36	331.93	15.16
Bedroom	Point 1	-23.00	346.00	-17.93	330.95	15.88
Bedroom	Point 1	-23.00	346.00	-18.50	330.25	16.38
Bedroom	Point 1	-23.00	346.00	-18.70	329.68	16.87
Bedroom	Point 1	-23.00	346.00	-18.47	329.45	17.16
Bedroom	Point 1	-23.00	346.00	-11.28	332.41	17.95
Bedroom	Point 1	-23.00	346.00	-11.48	331.93	18.18
Bedroom	Point 1	-23.00	346.00	-11.64	331.87	18.13
Bedroom	Point 1	-23.00	346.00	-12.12	331.36	18.24
Bedroom	Point 1	-23.00	346.00	-12.59	330.75	18.47
Bedroom	Point 2	14.00	472.00	16.60	453.20	18.98
Bedroom	Point 2	14.00	472.00	16.60	453.20	18.98
Bedroom	Point 2	14.00	472.00	16.60	453.20	18.98
Bedroom	Point 2	14.00	472.00	16.60	453.20	18.98
Bedroom	Point 2	14.00	472.00	16.60	453.20	18.98
Bedroom	Point 2	14.00	472.00	19.53	459.52	13.66
Bedroom	Point 2	14.00	472.00	17.58	458.38	14.08
Bedroom	Point 2	14.00	472.00	15.43	457.16	14.91
Bedroom	Point 2	14.00	472.00	14.69	456.17	15.85
Bedroom	Point 2	14.00	472.00	15.26	455.37	16.68

Table 22 shows that the average mean absolute error calculated for localization over 10 iterations for two points in the bedroom was 17.12 cm.

Table 23: Mean absolute error for points measured in the dining room area of HINT.

Area	Point	Actual X	Actual Y	Measured X	Measured Y	Error (in cm)
Dining room	Point 1	-238	259	-289.17	274.37	53.42
Dining room	Point 1	-238	259	-286.11	279.99	52.49
Dining room	Point 1	-238	259	-285.17	281.35	52.20
Dining room	Point 1	-238	259	-284.87	281.42	51.96
Dining room	Point 1	-238	259	-283.60	284.06	52.04
Dining room	Point 1	-238	259	-284.67	294.49	58.64
Dining room	Point 1	-238	259	-285.76	292.53	58.36
Dining room	Point 1	-238	259	-286.42	293.19	59.27
Dining room	Point 1	-238	259	-289.78	285.71	58.27
Dining room	Point 1	-238	259	-290.38	285.06	58.50
Dining room	Point 2	-238	155	-282.95	164.11	45.87
Dining room	Point 2	-238	155	-283.28	163.83	46.13
Dining room	Point 2	-238	155	-283.32	163.87	46.18
Dining room	Point 2	-238	155	-284.42	162.71	47.06
Dining room	Point 2	-238	155	-284.70	162.81	47.35
Dining room	Point 2	-238	155	-282.45	162.53	45.08
Dining room	Point 2	-238	155	-282.57	162.80	45.25
Dining room	Point 2	-238	155	-282.39	162.22	44.97
Dining room	Point 2	-238	155	-282.94	162.16	45.51
Dining room	Point 2	-238	155	-283.02	162.16	45.59

Table 23 shows that the average mean absolute error calculated for localization over 10 iterations for two points in the dining room was 50.70 cm.

Table 24: Mean absolute error for points measured in the entrance area of HINT.

Area	Point	Actual X	Actual Y	Measured X	Measured Y	Error (in cm)
Entrance	Point 1	194	171	220.11	131.20	47.60
Entrance	Point 1	194	171	220.04	131.15	47.60
Entrance	Point 1	194	171	219.74	130.86	47.68
Entrance	Point 1	194	171	219.31	131.33	47.06
Entrance	Point 1	194	171	219.17	131.40	46.92
Entrance	Point 1	194	171	214.64	139.65	37.53
Entrance	Point 1	194	171	214.85	139.84	37.49
Entrance	Point 1	194	171	215.85	140.69	37.36
Entrance	Point 1	194	171	216.11	140.16	37.95
Entrance	Point 1	194	171	216.29	139.89	38.27
Entrance	Point 2	281	276	332.16	215.46	79.26
Entrance	Point 2	281	276	333.53	213.76	81.44
Entrance	Point 2	281	276	334.98	212.04	83.69
Entrance	Point 2	281	276	336.69	209.41	86.80
Entrance	Point 2	281	276	337.13	207.97	88.19
Entrance	Point 2	281	276	325.04	219.22	71.85
Entrance	Point 2	281	276	324.45	219.96	70.91
Entrance	Point 2	281	276	324.33	219.89	70.89
Entrance	Point 2	281	276	323.75	220.14	70.34
Entrance	Point 2	281	276	324.55	219.78	71.11

Table 24 shows that the average mean absolute error calculated for localization over 10 iterations for two points at the entrance was 59.99 cm.

Table 25: Mean absolute error for points measured in the Kitchen area of HINT.

Area	Point	Actual X	Actual Y	Measured X	Measured Y	Error (in cm)
Kitchen	Point 1	-183	365	-161.86	360.63	21.58
Kitchen	Point 1	-183	365	-163.61	359.31	20.20
Kitchen	Point 1	-183	365	-164.22	359.16	19.67
Kitchen	Point 1	-183	365	-164.80	358.78	19.24
Kitchen	Point 1	-183	365	-164.34	361.76	18.94
Kitchen	Point 1	-183	365	-170.49	363.09	12.65
Kitchen	Point 1	-183	365	-172.35	362.82	10.87
Kitchen	Point 1	-183	365	-172.62	362.96	10.58
Kitchen	Point 1	-183	365	-171.24	364.23	11.78
Kitchen	Point 1	-183	365	-171.41	364.25	11.62
Kitchen	Point 2	-244	461	-225.58	456.81	18.89
Kitchen	Point 2	-244	461	-226.28	456.64	18.25
Kitchen	Point 2	-244	461	-226.18	456.79	18.31
Kitchen	Point 2	-244	461	-226.55	456.96	17.91
Kitchen	Point 2	-244	461	-225.82	458.22	18.39
Kitchen	Point 2	-244	461	-226.84	461.08	17.16
Kitchen	Point 2	-244	461	-226.38	461.27	17.62
Kitchen	Point 2	-244	461	-226.31	461.29	17.69
Kitchen	Point 2	-244	461	-226.08	461.42	17.92
Kitchen	Point 2	-244	461	-225.87	461.54	18.14

Table 25 shows that the average mean absolute error calculated for localization over 10 iterations for two points in the kitchen was 16.87 cm.

Table 26: Mean absolute error for points measured in the living room area of HINT.

Area	Point	Actual X	Actual Y	Measured X	Measured Y	Error (in cm)
Living room	Point 1	23.00	171.00	44.57	170.46	21.58
Living room	Point 1	23.00	171.00	30.11	179.57	11.14
Living room	Point 1	23.00	171.00	45.59	175.81	23.10
Living room	Point 1	23.00	171.00	23.85	177.48	6.53
Living room	Point 1	23.00	171.00	39.01	167.81	16.32
Living room	Point 1	23.00	171.00	25.16	177.07	6.44
Living room	Point 1	23.00	171.00	24.48	177.30	6.47
Living room	Point 1	23.00	171.00	38.23	179.45	17.42
Living room	Point 1	23.00	171.00	24.29	177.22	6.35
Living room	Point 1	23.00	171.00	30.85	175.89	9.25
Living room	Point 2	-94.00	259.00	-63.12	270.13	32.82
Living room	Point 2	-94.00	259.00	-64.59	268.93	31.04
Living room	Point 2	-94.00	259.00	-63.11	269.47	32.62
Living room	Point 2	-94.00	259.00	-59.59	270.59	36.31
Living room	Point 2	-94.00	259.00	-80.43	268.36	16.48
Living room	Point 2	-94.00	259.00	-81.24	268.47	15.90
Living room	Point 2	-94.00	259.00	-81.27	268.53	15.90
Living room	Point 2	-94.00	259.00	-83.14	269.06	14.81
Living room	Point 2	-94.00	259.00	-81.73	270.23	16.64
Living room	Point 2	-94.00	259.00	-70.97	272.15	26.52

Table 26 shows that the average mean absolute error calculated for localization over 10 iterations for two points in the living room was 18.18 cm.

Table 27: Falling forward (fall_f), Falling backward (fall_b), Falling sideways (fall_s), Fall detected (y), Fall not detected (n).

Action	Area	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5
fall_f	living	y	n	y	y	y
fall_b	living	y	y	y	y	y
fall_s	living	y	y	y	y	y
fall_f	kitchen	y	y	y	y	y
fall_b	kitchen	y	y	n	y	n
fall_s	kitchen	y	y	y	y	y
fall_f	bedroom	y	y	y	y	n
fall_b	bedroom	y	y	n	y	n
fall_s	bedroom	y	y	y	y	y
sitting	living	n	n	n	n	n
sitting	kitchen	n	n	n	n	n
sitting	bedroom	n	n	n	n	n
bending down	living	n	n	n	n	n
bending down	kitchen	n	n	y	n	n
bending down	bedroom	n	n	n	n	n
fall_s	dining	y	n	n	n	n
fall_s	entrance	n	n	n	n	n

The table shows the results of fall detection in different areas of the HINT. In the table, 'y' indicates that a fall was detected, and 'n' indicates that a fall was not detected in that iteration. This Table 27 supports the consolidated accuracy results presented in the result section.

Table 28: Intruder detected (y), Intruder not detected (n).

Intruder entering from	Action	Intruder going to	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5
main door	walking	living room	y	y	y	y	y
main door	walking	bedroom	y	y	y	y	y
main door	walking	kitchen	y	y	y	y	y
emergency exit	walking	living room	y	y	y	y	y
emergency exit	walking	bedroom	y	y	y	y	y
emergency exit	walking	kitchen	y	y	y	y	y
main door	crawling	living room	n	y	y	n	n
main door	crawling	bedroom	y	y	y	y	y
main door	crawling	kitchen	y	y	y	y	y
emergency exit	crawling	living room	n	n	y	n	n
emergency exit	crawling	bedroom	y	y	y	n	y
emergency exit	crawling	kitchen	n	y	y	y	n
main door	crouching	living room	y	y	y	y	y
main door	crouching	bedroom	y	y	y	y	y
main door	crouching	kitchen	y	y	y	y	y
emergency exit	crouching	living room	y	y	y	y	y
emergency exit	crouching	bedroom	y	y	y	y	y
emergency exit	crouching	kitchen	y	y	y	y	y

Table 28 shows the results of intruder detection from both entry points. In the table, 'y' indicates that an intruder was detected, and 'n' indicates that an intruder was not detected. Intruders walking, crawling, and crouching positions through the main door and emergency exit were simulated, with their intended destination being the living room, bedroom, or kitchen.

Table 29: Multiple people detected (y), Multiple people not detected (n).

Visitor ID	Visitor position	Resident position	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5
1	living room	living room	y	y	n	y	y
1	bedroom	living room	y	y	y	n	y
1	kitchen	living room	y	y	y	y	y
1	living room	bedroom	y	y	n	y	y
1	bedroom	bedroom	y	n	y	n	y
1	kitchen	bedroom	y	y	y	y	y
1	living room	kitchen	y	y	y	y	y
1	bedroom	kitchen	y	y	y	y	y
1	kitchen	kitchen	y	y	y	y	y
2	living room	living room	y	y	y	y	y
2	bedroom	living room	y	y	y	y	y
2	kitchen	living room	y	y	y	y	y
2	living room	bedroom	y	y	y	y	y
2	bedroom	bedroom	y	n	y	y	y
2	kitchen	bedroom	y	y	y	y	y
2	living room	kitchen	y	y	y	y	y
2	bedroom	kitchen	y	y	y	y	y
2	kitchen	kitchen	y	y	y	y	y
3	living room	living room	y	y	y	y	y
3	bedroom	living room	y	y	y	y	y
3	kitchen	living room	y	y	y	y	y
3	living room	bedroom	y	y	y	y	y
3	bedroom	bedroom	y	y	y	y	y
3	kitchen	bedroom	y	y	y	y	y
3	living room	kitchen	y	y	y	y	y
3	bedroom	kitchen	y	y	y	y	y
3	kitchen	kitchen	y	y	y	y	y

Table 29 shows the results of multiple people detection with three different visitors and a common resident. In the table, 'y' indicates that multiple people were detected, and 'n' indicates that multiple people were not detected. The combination of the visitor and resident in these three rooms (living room, bedroom, and kitchen) is considered for this tabulation. Additional cases where the visitor and resident are in a straight line relative to the radar sensor are also taken in Table 30 to address occlusion effects. Finally, all these 'y' and 'n' are consolidated to find the accuracy of the multiple people detection.

Table 30: Occlusion: Multiple people detected (y), Multiple people not detected (n).

Visitor ID	Visitor position	Resident position	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5
1	living room	living room	y	y	n	y	n
1	bedroom	bedroom	n	n	y	n	y
1	kitchen	kitchen	y	y	y	y	y
2	living room	living room	n	n	n	n	n
2	bedroom	bedroom	n	n	n	n	n
2	kitchen	kitchen	y	y	y	y	y
3	living room	living room	y	y	n	n	y
3	bedroom	bedroom	y	n	y	y	n
3	kitchen	kitchen	y	y	y	y	y

A.1 SURVEY

Table 31: Survey data Application weights

Application	Number of participants	Number of people agreed to the given ratings
Localization	59	46
Human Activity Recognition	59	57
Intruder detection	59	46
Fall detection	59	45
Multiple people detection	59	52
Visitor identification	59	49
Heart and breath rate detection	59	36

Table 31 shows the results of a survey conducted to support the assigned application weights in the benchmark model calculation. A total of 59 people participated in the survey, and the majority agreed with the application ratings provided.

Table 32 presents the assigned application weights for both accuracy and response time, along with the opposing viewpoints gathered from the survey. The numbers in the opposing viewpoints column indicate how many participants suggested modifying the given weights to the displayed alternative values.

Table 32: Assigned application weight and opposing viewpoints (Accuracy and Response Time)

Assigned application weight (Accuracy)	Opposing viewpoints (Accuracy)	Assigned application weight (Response time)	Opposing viewpoints (Response time)
(O)	4(++), 9(+)	(+)	8(++), 3(+), 2(-)
(+)	2(+)	(+)	2(++)
(++)	1(+)	(+)	13(++)
(++)	2(+)	(+)	14(++)
(+)	4(+)	(O)	2(++), 5(+)
(+)	3(++), 1(O)	(O)	4(++), 6(+)
(+)	15(++)	(O)	10(++), 13(++)

Table 33 provides the results of a survey on the feature score rating criteria for accuracy and response time. The accuracy for localization and heart & breath rate detection is measured differently from the other applications. In all cases, the majority of participants agreed with the proposed ratings.

Table 33: Feature score criteria survey details

Feature score	Number of participants	Number of people agreed to the given ratings
Accuracy localization	59	58
Accuracy heart and breath rate detection	59	59
Accuracy remaining application	59	59
Response time (for all applications)	59	58



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